Group 4 - Team 25

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From Prediction to Decision: Evaluating Travel Insurance Using Flight Cancellation and Delay Risk Models

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Abstract

Travel insurance is widely offered, yet its economic value at the time of booking is rarely evaluated using data-driven methods. We combine flight, fare, and weather data to predict delays and cancellations, translating predictions into expected value and break-even price thresholds. By ranking flights into risk percentiles, we estimate disruption probabilities, including missed connections.

Insurance is economically justified only for a small subset of high-risk flights, though this set expands as predictive performance improves. An application demonstrates how predictive models support transparent, economically grounded insurance decisions based on personalized price thresholds.

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Statement of Originality

We, the aforementioned students, herewith declare to have written this document and that we are responsible for the content of it. We declare that the text and the work presented in this document is original and that no sources other than those mentioned in the text and its references have been used in creating it.

Utrecht University School of Economics is responsible solely for the supervision of completion of the work, not for the content

Disclosure Statement

In this project, we have made use of the following Generative AI tools: Claude (Anthropic) and ChatGPT (OpenAI).

We have used these tools for: supporting the writing and coding process. This includes improving academic tone, sentence clarity and phrasing. AI was also used to help with executing our ideas, mainly those that went far beyond course material. Content regarding the creative process and main ideas behind the direction of our project is our own. Creative use of analyses, interpretation of results, and conclusions are mostly our own. However, AI-generated suggestions regarding these were reviewed and adapted when deemed an improvement on our own work by ourselves throughout.

Division of Work

We, the aforementioned students, herewith declare that all work on this project was carried out jointly and equally. All stages of the research, including data collection, analysis, and writing, were completed collaboratively by all group members.

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1. Introduction

Travel insurance is widely offered and frequently purchased, yet its economic value for travelers is rarely evaluated from the traveler's perspective using data-driven methods. While flight delays and cancellations are well-documented sources of uncertainty in air travel, it remains unclear whether available information allows travelers to make economically optimal insurance decisions at the time of booking, when most purchase decisions are made. In particular, the extent to which predictive models can improve such decisions and for whom insurance is actually worthwhile remains an open question.

Existing studies on flight disruption primarily focus on improving predictive accuracy (e.g., Gui et al., 2020; Chakrabarty et al., 2019), with limited attention to how predictions translate into economic decision-making. The literature on travel insurance, meanwhile, examines determinants of insurance purchase decisions based on traveler characteristics, but does not evaluate whether insurance is economically justified given flight-specific disruption risk. To our knowledge, existing research has not combined flight disruption prediction with a decision framework that evaluates the economic value of insurance at the level of individual flights.

This paper evaluates when travel insurance is economically justified for travelers at the time of booking by combining predictive models of flight disruptions with an expected value framework. Using flight operations, fare, and weather data for U.S. domestic flights between 2010 and 2019, we estimate cancellation and delay risks using numeric XGBoost and translate these predictions into economically grounded decision rules based on break-even thresholds. By ranking flights into risk percentiles, the approach captures how predictive performance translates into economically meaningful outcomes.

We estimate models under different information regimes to reflect the evolution of available information over time. The booking-time model uses only information available at the time of purchase and therefore represents the primary decision environment for travelers. A live-weather specification incorporates realized weather conditions at the day of departure, providing a benchmark for how improved information affects predictive performance and decision quality. In addition, we simulate models with higher predictive accuracy to assess how further improvements beyond observed data availability translate into economic outcomes.

The results show that insurance is economically justified only for a relatively small subset of high-risk flights, though this subset expands as predictive performance improves. The gains from better models differ across insurance types and connection window lengths. Cancellation insurance exhibits larger shifts in decision boundaries as predictive performance improves, while connection insurance is more sensitive to model quality for longer connection windows, where outcomes depend on rare but severe delays. Connection insurance applies to a broader set of flights, as disruption risk can be differentiated across a wider range of observations, whereas cancellation insurance is concentrated in a smaller high-risk segment. Together, these findings highlight that the value of predictive analytics depends not only on accuracy, but on how predictions are translated into decisions.

Methodologically, this paper contributes by translating machine learning-based risk predictions into a personalized expected value framework. Rather than evaluating models solely on metrics like AUC, we calculate dynamic break-even price thresholds for individual flights. By comparing models under booking-time and live-weather information sets, we explicitly quantify the economic value of information timing. Furthermore, by incorporating Signal Detection Theory simulations and an interactive decision-support application, we provide a direct assessment of the practical economic value of predictive models in a real-world context.

The remainder of the paper is structured as follows. Section 2 describes the data sources and preparation. Section 3 presents descriptive analyses of disruption patterns. Section 4 outlines the empirical approach and modeling framework. Section 5 presents the main results, including decision outcomes and expected value implications. Section 6 concludes.

2. Data

Our analysis relies on flight-level information on departure performance and route characteristics from the Bureau of Transportation Statistics (BTS, n.d.), fare and passenger demand data from the BTS Origin and Destination Survey (DB1B), and daily weather observations from the NOAA Global Surface Summary of Day (GSOD) (NCEI, n.d.). These three sources were merged into a single dataset covering US domestic flights from 2010 to 2019.

2.1 Flight Performance Data (Kaggle, Carrier On-Time Performance)

The base dataset was sourced from Kaggle (Maxwell, n.d.), originating from the Bureau of Transportation Statistics (BTS), and contains 2,000,000 flight-level observations from 1987 to 2020. The dataset was filtered to 2010 to 2019 to avoid structural disruptions from events such as the September 11 attacks, the 2008/2009 financial crisis, and the COVID-19 pandemic, which would introduce patterns unrepresentative of normal operating conditions. Columns that were entirely or almost entirely missing were also removed, primarily those related to diverted flight operations, which are too rare to contribute meaningfully to delay prediction.

2.2 Fare Data (Bureau of Transportation Statistics, DB1B)

To investigate whether fare prices are correlated with delay risk and flight cancellations, quarterly fare data from the BTS Origin and Destination Survey (DB1B) was added as an exploratory predictor for the same 2010 to 2019 period. The DB1B is a mandatory 10% random sample of domestic tickets reported quarterly by carriers (BTS, n.d.). Gerardi and Shapiro (2009) use the same DB1B data source and demonstrate that route-level fare variation across US domestic markets is systematically related to market structure and competition, validating the DB1B as a reliable measure of route-level pricing and motivating its inclusion as a predictor variable in our delay models. A passenger weighted average fare was computed per route, carrier, year, and quarter to ensure high volume bookings contribute proportionally to the route level estimate. Following Gerardi and Shapiro (2009), fares below $25 were excluded using a slightly more conservative threshold than their $20 cutoff, for the same reasons of removing suspected data errors and frequent flyer redemptions. Fares above $1,100 were additionally excluded to remove extreme outliers that would otherwise upwardly skew average fare calculations. After filtering and merging, the dataset contained 635,311 flight records. Further details on the merging procedure are provided in Appendix A.

2.3 Weather Data (NOAA GSOD)

Weather conditions at both departure and arrival airports were sourced from the NOAA Global Surface Summary of Day (GSOD) dataset (NCEI, n.d.). Khaksar and Sheikholeslami (2016) find that visibility, wind speed, and departure time are among the most influential predictors of departure delay in the US airline network, directly motivating the inclusion of these weather variables in our feature set. Station metadata was obtained from the NOAA Integrated Surface Database (ISD) station history file (NCEI, n.d.), from which 2,448 US stations with valid coordinates and continuous coverage over 2010 to 2019 were retained. Each airport was matched to its nearest station using the Haversine formula, with airports more than 30km from any station excluded due to potential inaccuracy in local weather representation, leaving 370 airports. Flights missing origin or destination temperature after merging were removed rather than imputed to avoid introducing systematic bias, resulting in a final dataset of 635,311 flights. Further details on variable construction and data cleaning are provided in Appendix A.

3. Descriptive Data Analysis

The two outcome variables show substantial class imbalance. Of 635,311 flights, 18.3% experienced an arrival delay of ≥15 minutes and 1.6% were cancelled, this a ratio of roughly 1:61 for cancellations. A classifier predicting the majority class would achieve 98.4% accuracy while providing no useful information. We therefore evaluate model performance using AUC, which measures ranking ability across all thresholds and is invariant to class imbalance.

Delay rates show clear temporal structure. Delay rates rise from approximately 10% for pre-07:00 departures to over 30% for post-20:00 flights. Clear seasonal patterns exist too. Delay and cancellation rates peak in summer and winter and reach their minimum in September–October. (See appendix J)

Airline and airport identity show structural differences in disruption frequency. Delay and cancellation rates vary substantially across carriers and airports, with major hub airports such as ORD, EWR, and JFK exhibiting systematically higher rates than smaller regional airports. (See appendix J)

Weather variables contribute 18 features covering temperature, wind, precipitation, snow depth, and visibility at both origin and destination. Precipitation and snow depth are zero-inflated, with most disruption concentrated in a small number of high-severity days. Visibility is bounded at 10 miles per aviation reporting conventions. In the booking-time model, all weather variables enter as monthly historical averages rather than realized values, which will lead to substantially reducing their predictive contribution relative to the live-weather specification.

4. Empirical Approach

We estimate disruption risk using a classification framework that predicts the probability of flight delays and cancellations. Separate models are estimated for delay risk (used for connection insurance) and cancellation risk (used for cancellation insurance), reflecting the distinct nature of these insurance products and their underlying decision problems.

4.1 Prediction Framework

For connection insurance, we model the probability of an arrival delay of at least 15 minutes (delay ≥ 15 minutes).[1] This threshold is chosen to balance predictive relevance and statistical stability. While missed connections are driven by more severe delays (e.g., ≥ 60 minutes), directly predicting such events leads to substantial class imbalance and reduced model performance. Using the ≥15-minute threshold allows the model to produce a more stable and informative ranking of flights by general disruption risk.

For cancellation insurance, we directly predict the probability that a flight is cancelled, as this outcome aligns exactly with the insurance event of interest.

Models are trained on 2010–2018 data and evaluated in 2019 to ensure a realistic out-of-sample setting while using the most recent stable data. This means AUC mentioned in our paper is test AUC.

Predictions are generated using gradient boosted decision trees implemented via XGBoost, which effectively capture nonlinear relationships and interactions across flight characteristics, routing variables, and weather conditions. To address class imbalance, we apply class weighting through the scale_pos_weight parameter. This improves the model's ability to correctly rank rare disruption events by penalizing misclassification of the minority class more heavily.

While this approach typically distorts the calibration of predicted probabilities, this does not affect our analysis. The framework relies on the relative ranking of flights rather than on absolute probability estimates. Since predictions are converted into percentile-based risk tiers, only the ordering of observations is relevant, and improved ranking performance directly translates into better identification of high-risk flights.

The objective of the analysis is predictive rather than causal, focusing on risk estimation rather than identification of structural relationships.

For interpretability, individual features are grouped into broader categories (e.g., time and schedule, weather conditions, route characteristics), allowing for a more concise comparison of model drivers across specifications.

We also evaluate alternative model specifications as robustness checks, including CatBoost, Lasso regression and a categorical implementation of XGBoost (Appendix B) and a cascade approach incorporating predicted departure delay probabilities(Appendix E). Only some of these models yield marginal improvements in predictive performance. Importantly, these differences do not materially affect the ranking of high-risk flights or the resulting insurance decisions. Decision boundaries and expected value outcomes remain largely unchanged across specifications, as documented in Appendix C. Given the negligible performance differences, we proceeded to keep the numeric XGBoost specification as our baseline model (Appendix B). We estimate two model specifications that differ in information availability:

- **Booking-time model**: uses only information available at the time of booking, including historical monthly weather averages. We specifically adopt monthly weather aggregation for this specification after evaluating more granular temporal alternatives, as documented in Appendix G
- **Live-weather model**: incorporates realized weather conditions on the day of departure.

The booking-time model reflects the information set available at the moment when most insurance purchase decisions are made. As such, it represents the primary decision environment for travelers. The live-weather model serves as a benchmark to assess how improvements in information availability affect risk prediction and economic outcomes.

Model performance is evaluated using the area under the ROC curve (AUC), with moderate improvements observed when incorporating real-time weather information.

4.2 Risk Ranking and Tier Construction

Flights are ranked based on predicted risk and grouped into percentile-based risk tiers (e.g., top 1%, top 10%). These tiers are constructed separately for each model to ensure comparability within each prediction setting. This ranking approach is central to the analysis, as insurance decisions depend on how well models prioritize high-risk flights rather than on absolute probability levels.

4.3 Mapping Predictions to Realized Outcomes

For connection insurance, predicted delay risk must be translated into the probability of missing a connection. Because the model predicts delays ≥15 minutes rather than extreme delays, we recover delay severity using the empirical distribution of realized delays within each risk tier.

Specifically, for each combination of risk tier and connection time window, we compute the probability that realized delays exceed the available connection time. This allows us to estimate missed connection probabilities despite using a lower prediction threshold.

For cancellation insurance, this step is more direct: within each risk tier, we compute the observed cancellation rate, which serves as the empirical probability of disruption.

4.4 Decision Framework

Insurance decisions are evaluated at the level of risk tiers and, for connection insurance, connection time windows. For each scenario, we compare the empirical disruption probability to the insurance premium to determine whether purchasing insurance is economically justified for a traveler facing a given itinerary.

- **Connection insurance**: expected value is based on the probability of a missed connection for each combination of risk tier and connection time window.
- **Cancellation insurance**: expected value is based on the average cancellation rate within each risk tier.

These probabilities are compared to the insurance premium, expressed as a percentage of ticket price. We assume premiums of 4.5% for connection insurance and 6.5% for cancellation insurance, consistent with the industry benchmark range of 4%–10% of total trip cost (InsureMyTrip, n.d.). The higher premium for cancellation insurance reflects the broader scope of coverage and greater potential payout compared to connection insurance, which covers a narrower and more predictable disruption event. Insurance is considered economically justified when the disruption probability exceeds the premium, with a small interval around zero treated as a break-even region.

These results should be interpreted as decision benchmarks rather than prescriptive recommendations, as optimal choices depend on individual risk preferences.

Expected value is expressed as a percentage of ticket price, allowing the framework to scale directly with individual itineraries. This also allows us to compute a break-even premium for each scenario, defined as the maximum insurance price at which a traveler is indifferent between purchasing and not purchasing insurance, given their estimated disruption risk.

The framework assumes risk-neutral decision-making. Under risk aversion, travelers' willingness to pay for insurance exceeds the expected loss, implying that insurance may be preferred even when expected value is slightly negative. For this reason, the analysis focuses on break-even price thresholds rather than binary purchase recommendations. These thresholds provide a flexible decision rule that allows travelers to incorporate individual risk preferences when evaluating insurance offers.

The decision problem can be framed as a comparison between the expected utility of purchasing insurance and remaining uninsured, where the risk-neutral case reduces to the expected value comparison used in this analysis.

4.5 Robustness and Model Evaluation

To ensure that the empirical delay distributions and resulting decision rules are stable, we evaluate the framework across multiple out-of-sample years. This verifies that the relationship between predicted risk tiers and realized disruption outcomes is not driven by a specific sample period.

In addition, we simulate models with higher predictive performance to assess how improvements in ranking quality affect decision-making. This allows us to distinguish between changes that shift decision boundaries and those that primarily increase expected value within already high-risk groups (see Appendix H for the simulation methodology grounded in Signal Detection Theory)

Footnote

[1] A natural alternative is to model delay magnitude directly using a regression framework. However, regression models perform poorly in this setting due to the high variability and skewness of delay outcomes, yielding predictions with low correlation to realized delays and substantial compression toward the mean(Appendix E). Since insurance outcomes depend on whether delays exceed specific thresholds rather than their exact magnitude, a classification-based approach combined with empirical delay distributions within risk tiers is more appropriate for the decision problem considered here..

5. Results

5.1 Predictive Performance

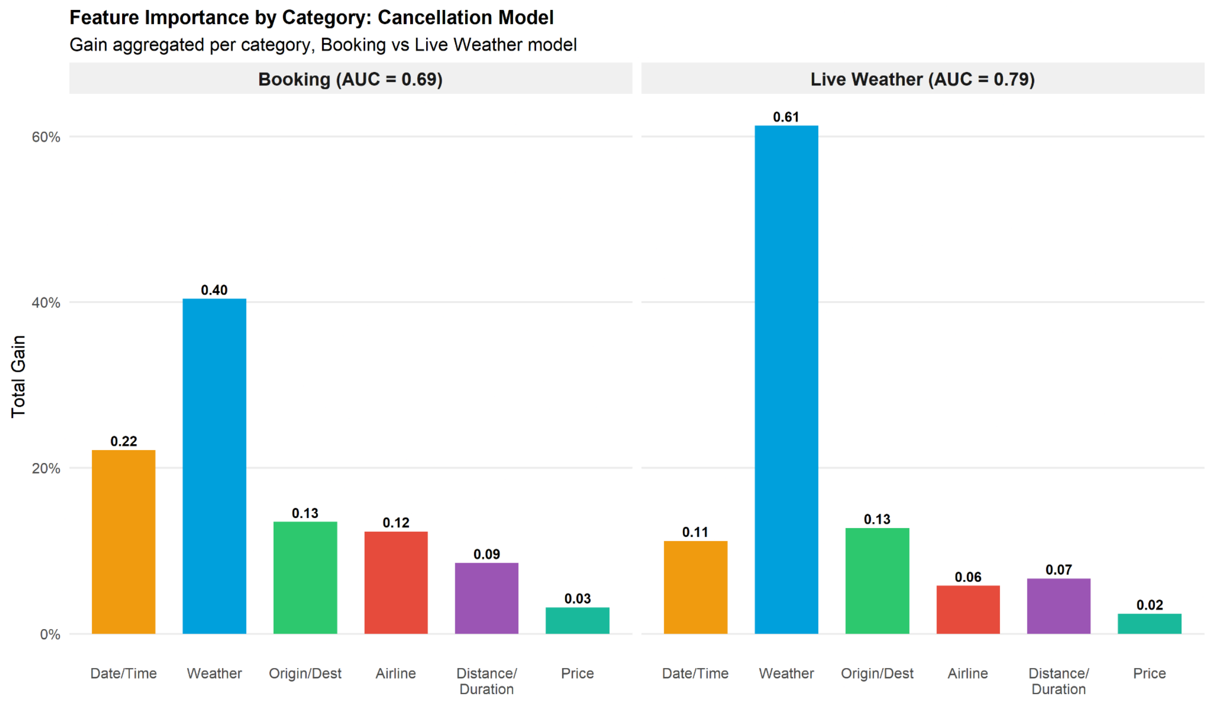

Figure 1: *Aggregated Cancellation Model XGBoost gain scores by feature category for the booking-time model (AUC = 0.69) and live weather model (AUC = 0.79), trained on U.S. domestic flight data (2010–2018).*

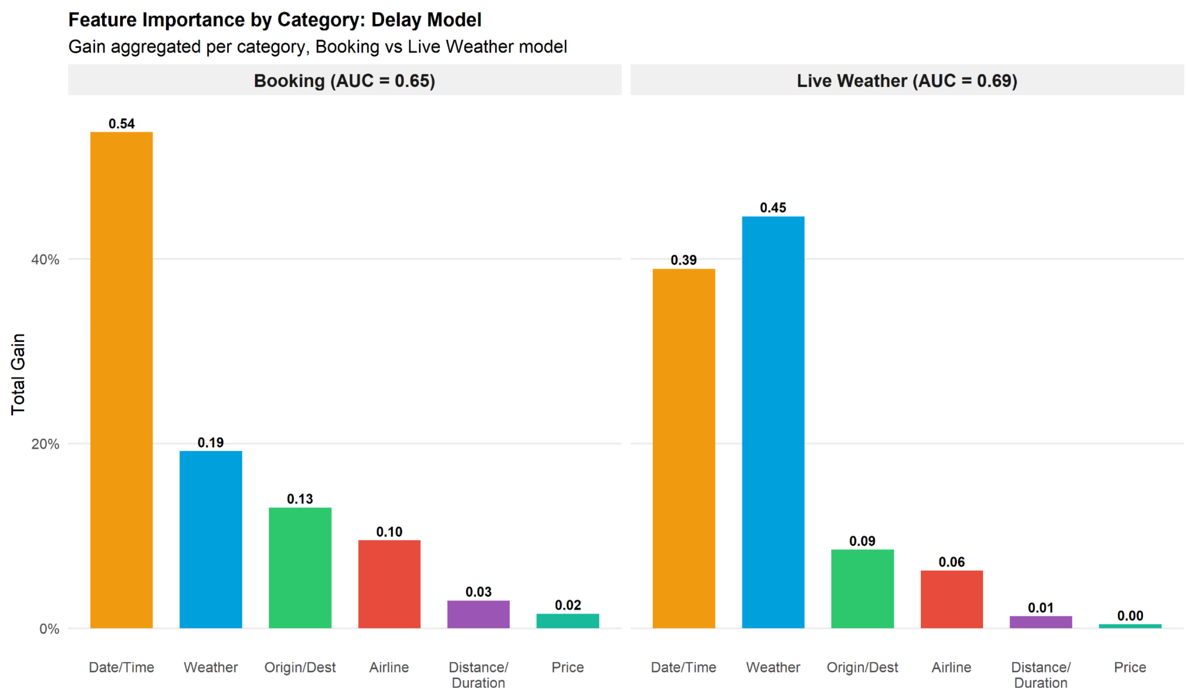

Figure 2: *Aggregated arrival delay XGBoost gain scores by feature category for the booking-time model (AUC = 0.65) and live weather model (AUC = 0.69), trained on U.S. domestic flight data (2010–2018).*

Delay Model

At booking time, Date/Time features dominate (0.54)[1], with scheduled departure time being the single most important predictor. This reflects that delay risk is largely structural — certain times of day, days of the week, and months are systematically more prone to delays regardless of conditions. Weather contributes modestly (0.19), which is expected given that booking-time weather relies on monthly averages rather than actual conditions. When live weather is incorporated, Date/Time drops to 0.39 while Weather rises to 0.45, becoming the dominant category. This shift explains the AUC improvement from 0.65 to 0.69, as real-time conditions add meaningful signal beyond structural patterns.

Cancellation Model

A similar but more pronounced pattern emerges. Weather already dominates at booking time (0.40 vs 0.22 for Date/Time), suggesting cancellations are more weather-driven than delays even at the structural level. With live weather, this gap widens dramatically, with Weather rising to 0.61 while Date/Time drops to 0.11. This explains the larger AUC gain for cancellations (0.69 → 0.79) relative to delays (0.65 → 0.69), as cancellations are fundamentally triggered by severe weather events, making real-time conditions far more informative.

Role of Ticket Price

Average fare contributes negligibly across all models and both outcomes (≤0.03), suggesting that route pricing carries little independent information about disruption risk beyond what is already captured by route, airline, and timing features. While fare data was included as a novel addition to the feature set to explore whether route-level pricing is associated with on-time performance, its marginal predictive contribution is minimal.

[1] Disaggregated feature importance scores for all individual features across both models and specifications are reported in Appendix F.

5.2 Connection Insurance: Decision Framework

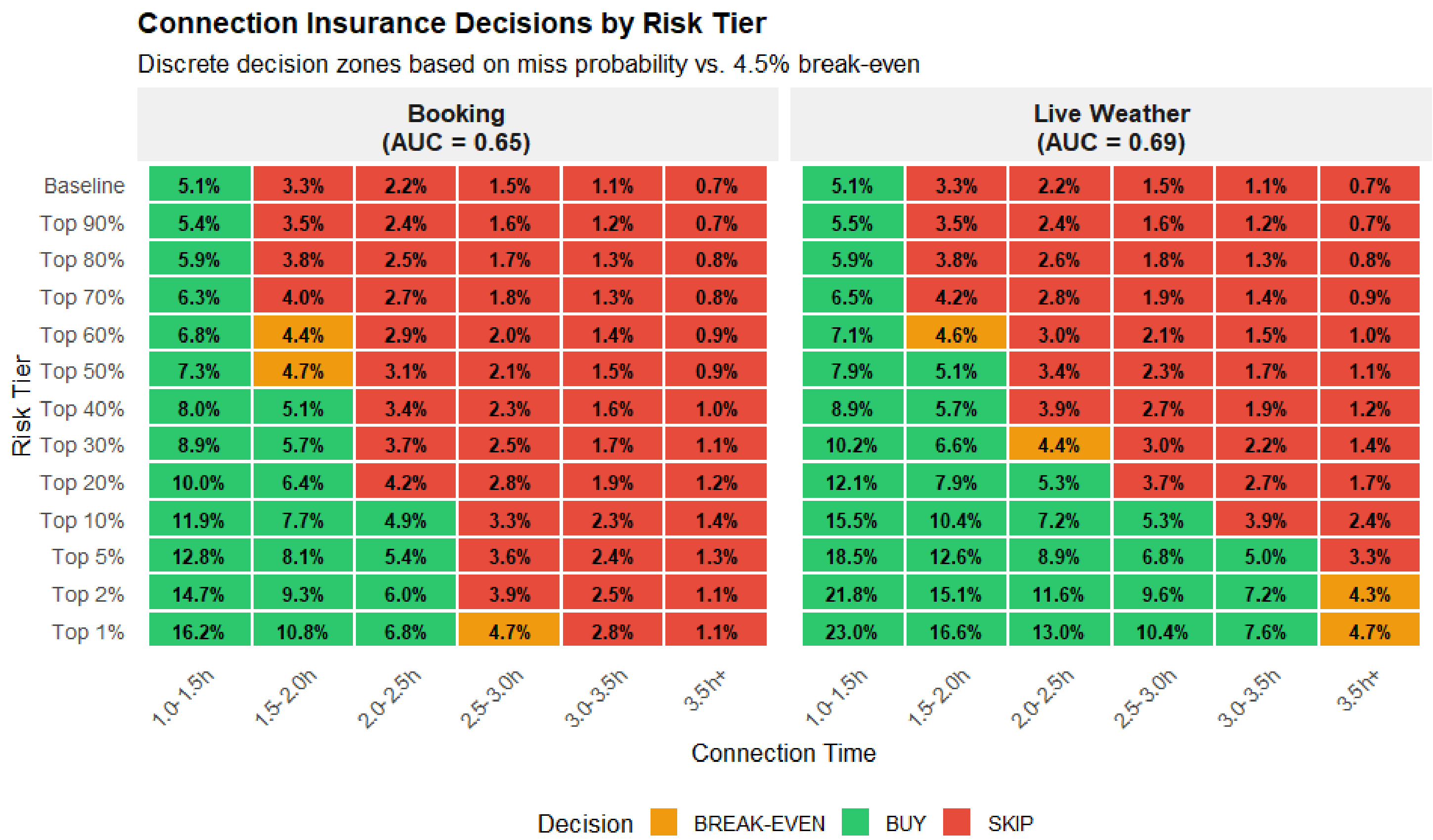

Figure 3 :*Connection Insurance Decision Matrix.* *This heatmap illustrates discrete decision zones for connection insurance based on the empirical probability of a missed connection versus a 4.5% premium break-even threshold. The matrix compares the risk-ranking performance of the Booking-time model (AUC = 0.65) against the Live Weather model (AUC = 0.69) across varying risk percentiles and connection time windows.*

Economic Value of Improved Risk Ranking

Despite only moderate predictive performance (AUC ≈ 0.69), the live-weather model provides clear economic value through improved risk ranking. Relative to the booking-time model, it more effectively concentrates high-risk flights in the upper percentiles. This is visible in the decision heatmap, where miss rates increase more sharply across risk tiers, resulting in a larger set of flights exceeding the break-even threshold for insurance. The improvement reflects better ordering of risk rather than higher overall predicted risk levels.

Variation by Connection Time

The differences between the models vary with connection time. For short connections (1–2 hours), both models identify similar high-risk groups. In this range, even moderate delays are sufficient to cause missed connections, so improvements in ranking have limited impact on decisions.

Ranking Breakdown at Long Connection Windows

For longer connection windows (≥3.5 hours), the models diverge. Missed connections are driven by relatively rare but severe delays, making accurate identification of extreme events more important. In this setting, the booking-time model performs poorly: miss rates do not increase monotonically and begin to converge across risk groups, particularly in the upper percentiles. For example, the top 1% group does not consistently exhibit higher miss rates than broader groups such as the top 20%, indicating a breakdown in ranking performance.

Live-Weather Model Consistency

In contrast, the live-weather model maintains a consistent ordering, with higher risk tiers corresponding to higher miss probabilities even at longer connection times. This suggests that the main limitation of the booking-time model is not its average predictive performance, but its inability to correctly rank extreme disruption risk.

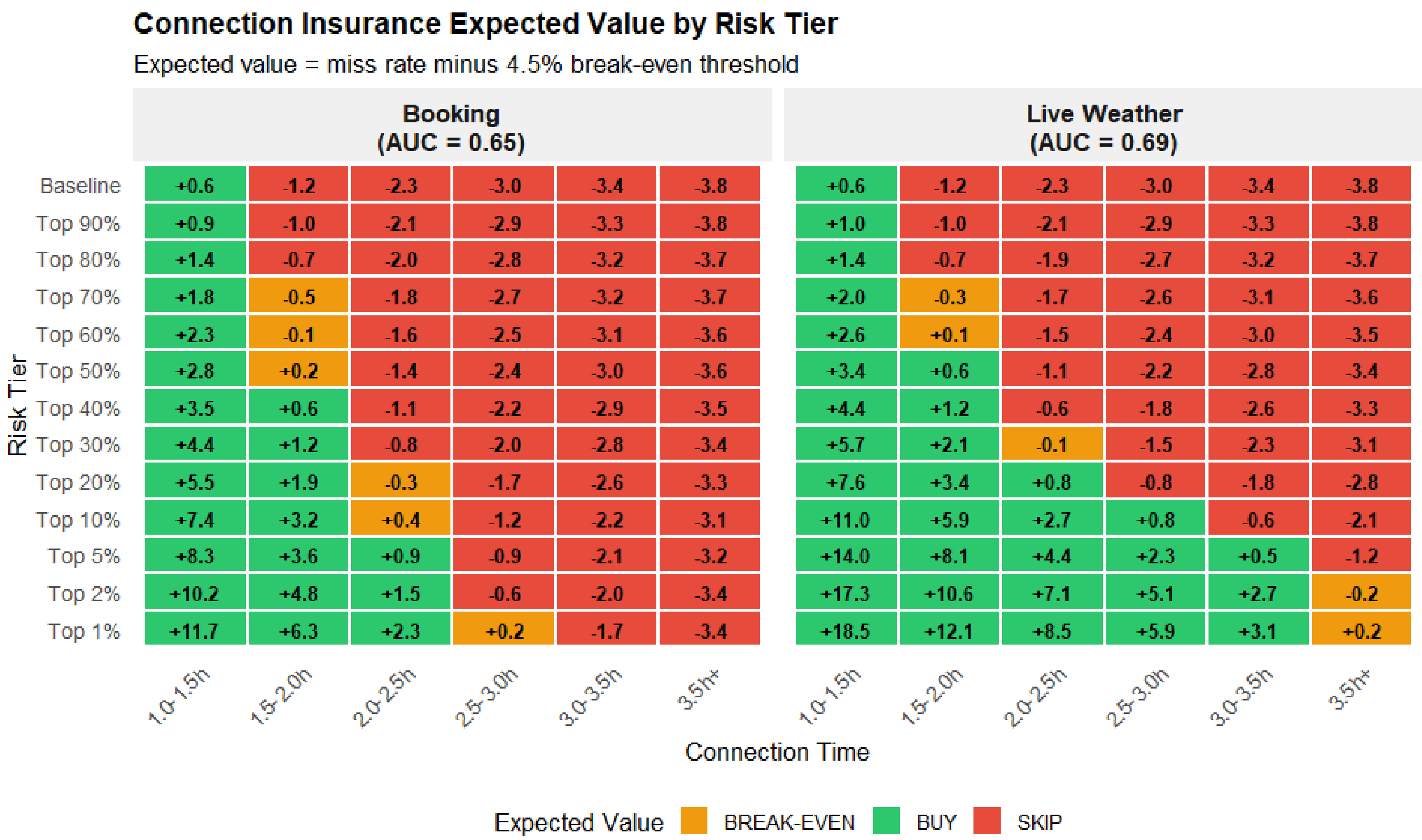

Figure 4: *Connection Insurance Decision Matrix***.** *Comparison of insurance recommendations between Booking-time (AUC = 0.65) and Live Weather (AUC = 0.69) models across risk percentiles and connection windows. Zones indicate whether the miss probability exceeds the 4.5% premium break-even threshold .*

Expected Value Implications

The expected value results show that improvements in ranking translate directly into economic gains. The live-weather model increases expected value in higher-risk tiers, converting marginal or negative outcomes under the booking-time model into clearly positive ones. This reflects improved concentration of disruption risk within the upper percentiles.

Effect of Connection Time on Economic Gains

The magnitude of these gains depends on connection time. For short connections, decision boundaries remain largely unchanged across models. Improvements in predictive performance mainly affect the level of expected value rather than the direction of decisions. For longer connection windows, the effect is stronger. Since missed connections depend on extreme delays, improved ranking materially affects both decisions and expected value. The live-weather model identifies a subset of flights where insurance becomes economically justified, while the booking-time model fails to isolate these cases.

Concentration of Gains in the Upper Tail

Across both figures, the gains from improved prediction are concentrated in the upper tail of the risk distribution. Changes in overall model performance have limited impact unless they improve the identification of rare, high-impact events.

Assumptions and Limitations

Results should be interpreted under the assumption of risk-neutral decision-making. In practice, risk-averse travelers may prefer insurance even when expected value is slightly negative.

Temporal Stability

The main patterns are stable across additional test years, with similar ranking behavior and expected value outcomes observed out-of-sample (see Appendix Figure C).

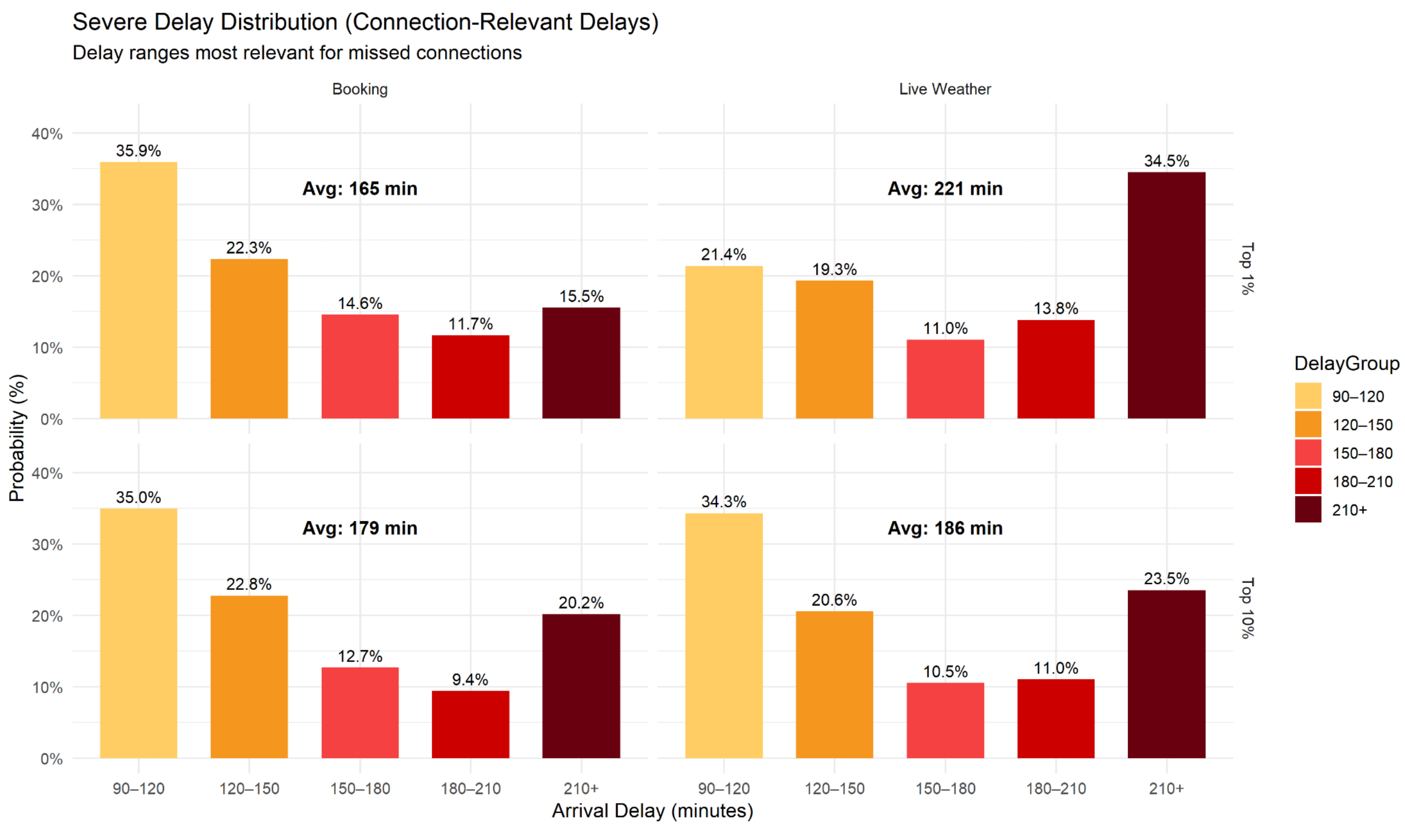

Figure 5: *Severe Delay Distribution by Risk Tier***.** *This chart compares how arrival delay severity is distributed within the Top 1% and Top 10% risk groups for both models. The Live Weather model demonstrates superior risk concentration by shifting probability mass toward the upper tail (210+ minutes) in the highest-risk tier. This illustrates the model's improved ability to identify weather-driven extreme delays most critical for connection insurance.*

Delay Severity Distribution Across Risk Tiers

To understand why the live-weather model improves decision outcomes, we examine how delay severity is distributed across risk tiers. Figure 3 reports the distribution of arrival delays within the top 1% and top 10% risk groups.

Short Connection Times

For short connection times (up to approximately 2 hours), both models exhibit similar delay distributions. This is consistent with earlier results, as even moderate delays are sufficient to cause missed connections, limiting the impact of improved ranking.

Divergence at Longer Connection Windows

For longer connection windows, the models diverge. In this setting, missed connections are driven by relatively rare but severe delays, which are largely associated with adverse weather conditions. Live weather data improves the identification of these severe, weather-driven delays by shifting probability mass toward the upper tail and concentrating extreme delays in the highest risk percentiles. As a result, high-risk flights correspond more consistently to severe outcomes.

Limitations of the Booking-Time Model

In contrast, the booking-time model relies on historical averages and therefore fails to capture real-time weather shocks. This leads to an overrepresentation of moderate delays and weaker concentration of extreme events in the top percentiles.

Implications for Insurance Decisions

Since missed connections for longer connection windows are driven by these severe delays, improved identification of weather-driven disruptions directly translates into more accurate insurance decisions.

5.3 Cancellation Insurance: Decision Framework

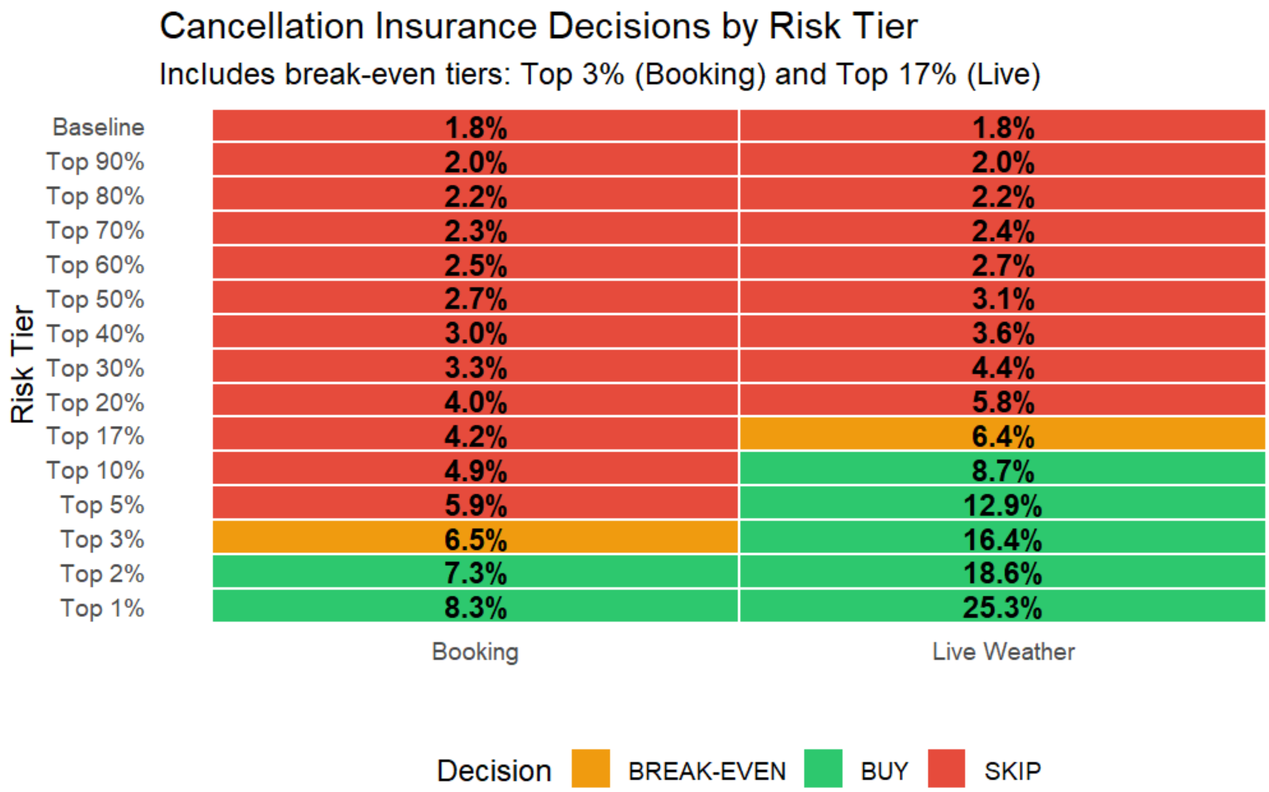

Figure 6: *Cancellation Insurance Decisions. This matrix compares the empirical decision boundaries of the Booking-time(AUC = 0.69) and Live Weather model(AUC = 0.79), showing the "BUY" region expanding from the Top 3% to the Top 17% as predictive accuracy improves.*

Cancellation Insurance: Model Comparison

A similar pattern emerges for cancellation insurance, though the differences across models are more pronounced. The key distinction lies in the ability to concentrate risk in the upper percentiles.

Risk Concentration and Break-Even Thresholds

Under the booking-time model, cancellation risk increases gradually across risk tiers and reaches the break-even threshold only at the top 3% of flights, indicating limited separation between moderate and high-risk observations. In contrast, the live-weather model reaches break-even much earlier, around the top 17%, and exhibits a much steeper increase in risk beyond this point.

Implications of Improved Ranking

This sharper gradient reflects improved ranking performance, where high-risk flights are more effectively concentrated in the upper percentiles. As a result, the set of flights for which insurance is economically justified expands substantially when live information is incorporated.

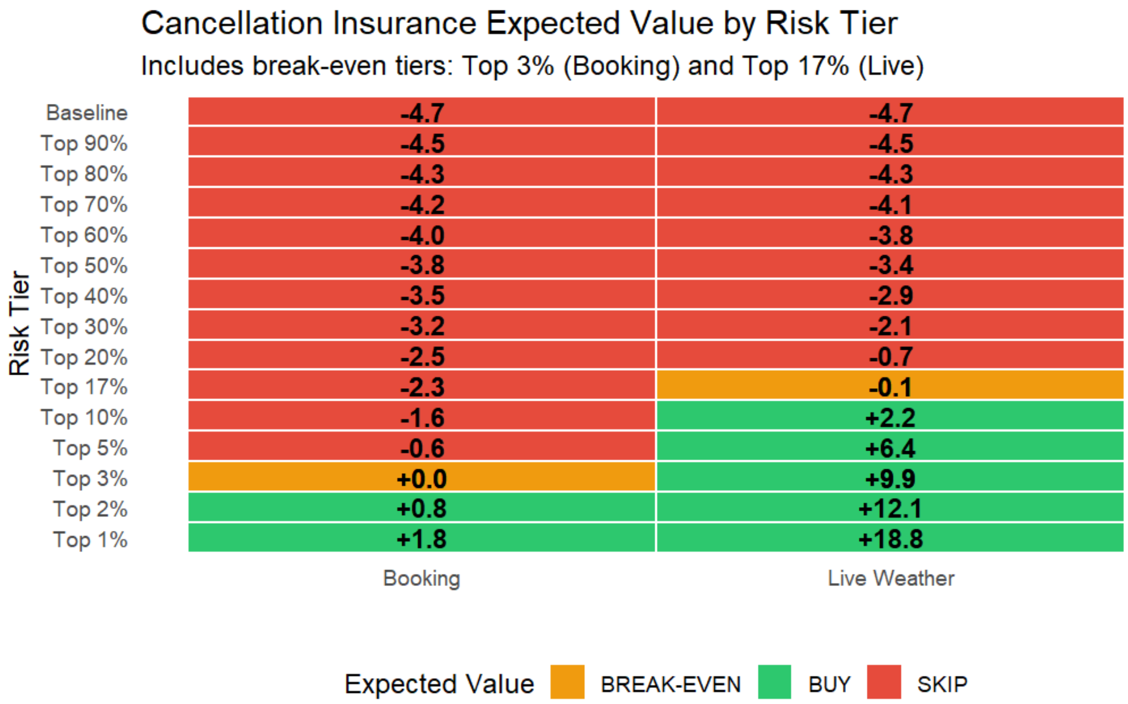

Figure 7: *Cancellation Insurance Expected Value. This matrix displays the net economic return (Cancel Rate - 6.5%) for each risk tier, showing how live weather data increases the maximum expected value from +1.8% to +18.8% of the ticket price.*

Expected Value Comparison

The expected value results reinforce this pattern. Under the booking-time model, expected value remains negative across most risk tiers and becomes only marginally positive in the extreme upper tail. In contrast, the live-weather model generates strongly positive expected values across a broader set of high-risk flights.

Risk Concentration as the Driver of Improvement

This improvement is driven by better risk concentration rather than higher average cancellation rates. By reallocating cancellation risk toward the upper percentiles, the model increases the average disruption probability within these groups, leading to more favorable insurance decisions.

Convexity of Gains in the Upper Tail

The gains are particularly pronounced in the highest-risk tiers, where expected value increases rapidly. This reflects a convex relationship between disruption probability and economic return, where improvements in identifying high-risk flights translate into disproportionately large gains in expected value.

5.4 Model Improvement & Simulation: how models compare to optimal models are they applicable to help most travelers?

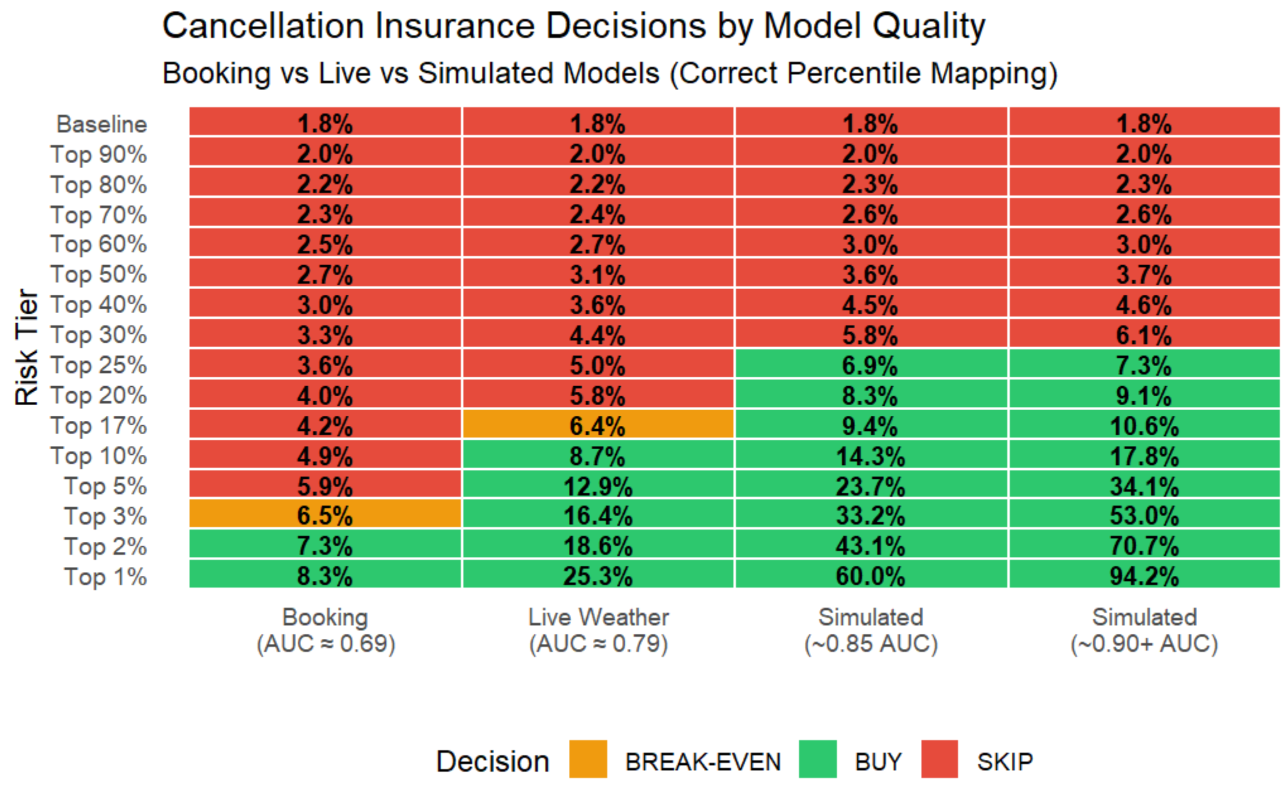

Figure 8: *Cancellation Insurance Sensitivity Analysis. This matrix illustrates how increasing model quality (AUC) shifts the 6.5% economic break-even point from the Top 3% in the booking model to the Top 25% in high-accuracy simulations, while simultaneously concentrating cancellation risk as high as 94.2% in the top percentile.*

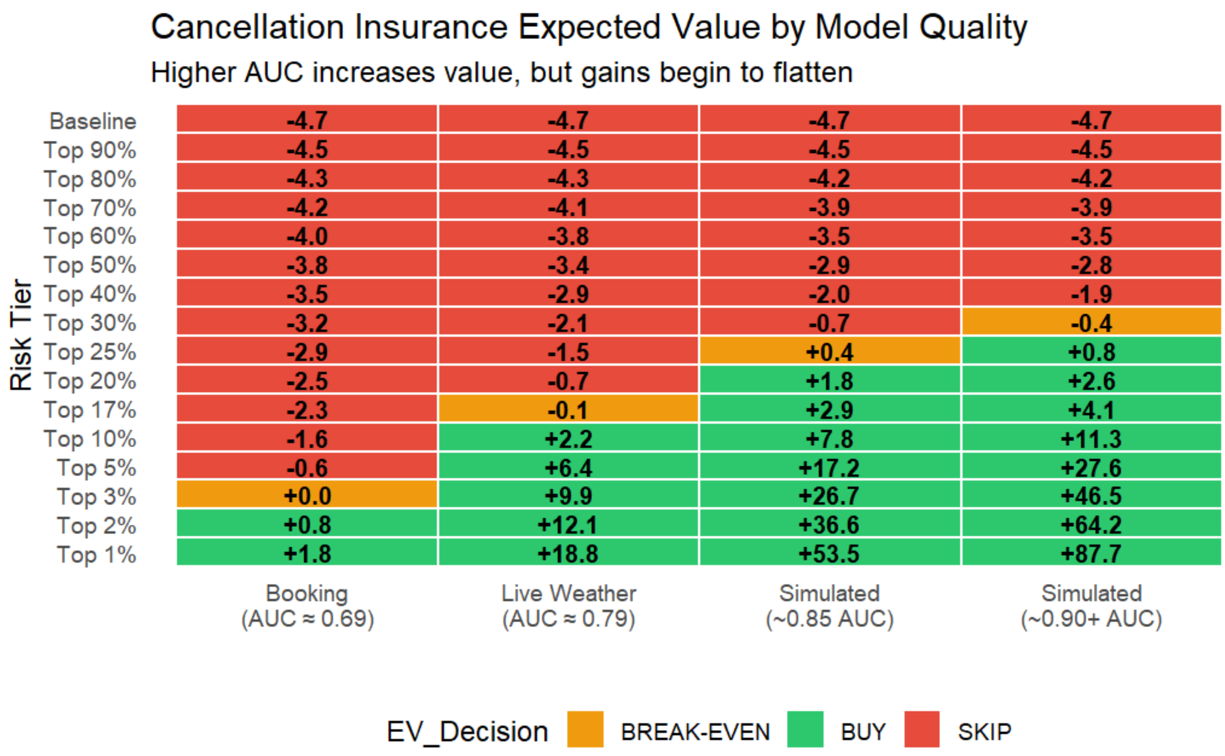

Figure 9: *Cancellation Insurance Expected Value (EV) Simulation Analysis. This matrix demonstrates that as model accuracy (AUC) increases, the potential net economic return per ticket scales dramatically, reaching as high as +87.7% in high-performance simulations within the top risk percentile.*

Effect of Increasing Model Quality

To generalize these findings, we compare models of increasing predictive quality. The results show that improvements in model performance primarily affect the concentration of risk in the upper tail of the distribution.

Expansion of the Decision Boundary

The booking-time model exhibits limited separation across risk tiers, with only a very small set of flights exceeding the break-even threshold. The live-weather model substantially improves this, shifting a larger share of flights above the threshold. As model quality increases further, the decision boundary expands to approximately the top 25 to 30 percent of flights.

Diminishing Returns to Decision Expansion

However, the largest gains occur in expected value rather than in the expansion of the decision set. Improvements in predictive performance primarily increase expected value within already high-risk groups, rather than substantially increasing the number of flights for which insurance is economically justified.

Practical Implications

This pattern indicates that most practical value for travelers is achieved once a sufficiently strong ranking is in place. Beyond that point, further improvements mainly increase the magnitude of returns in the highest-risk tiers rather than changing the underlying decision.

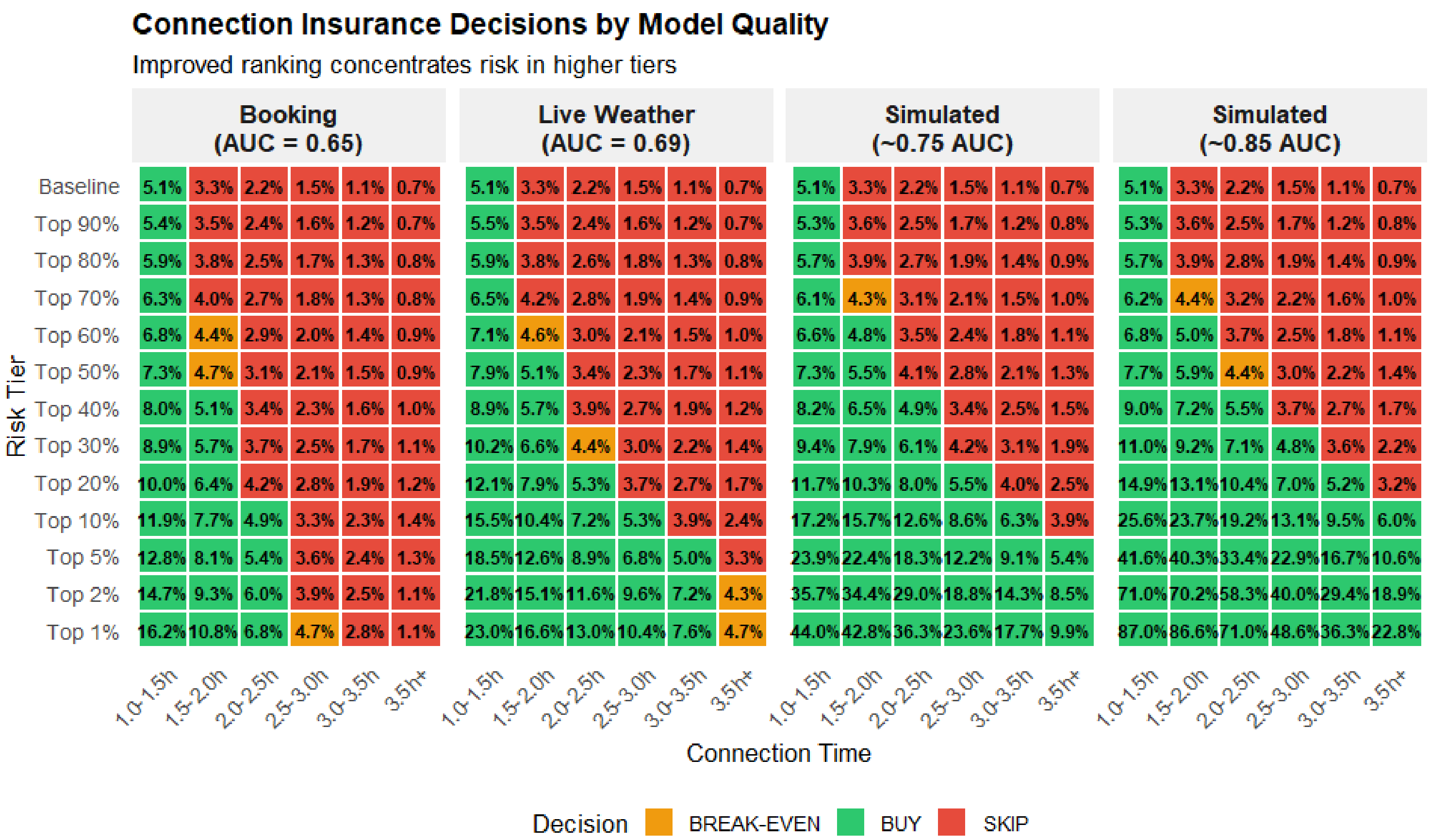

Figure 10: *Connection Insurance Decision Sensitivity Analysis. This matrix illustrates that higher model accuracy (AUC) significantly expands the economically viable "BUY" region across wider connection windows while simultaneously intensifying the concentration of risk within the highest tiers.*

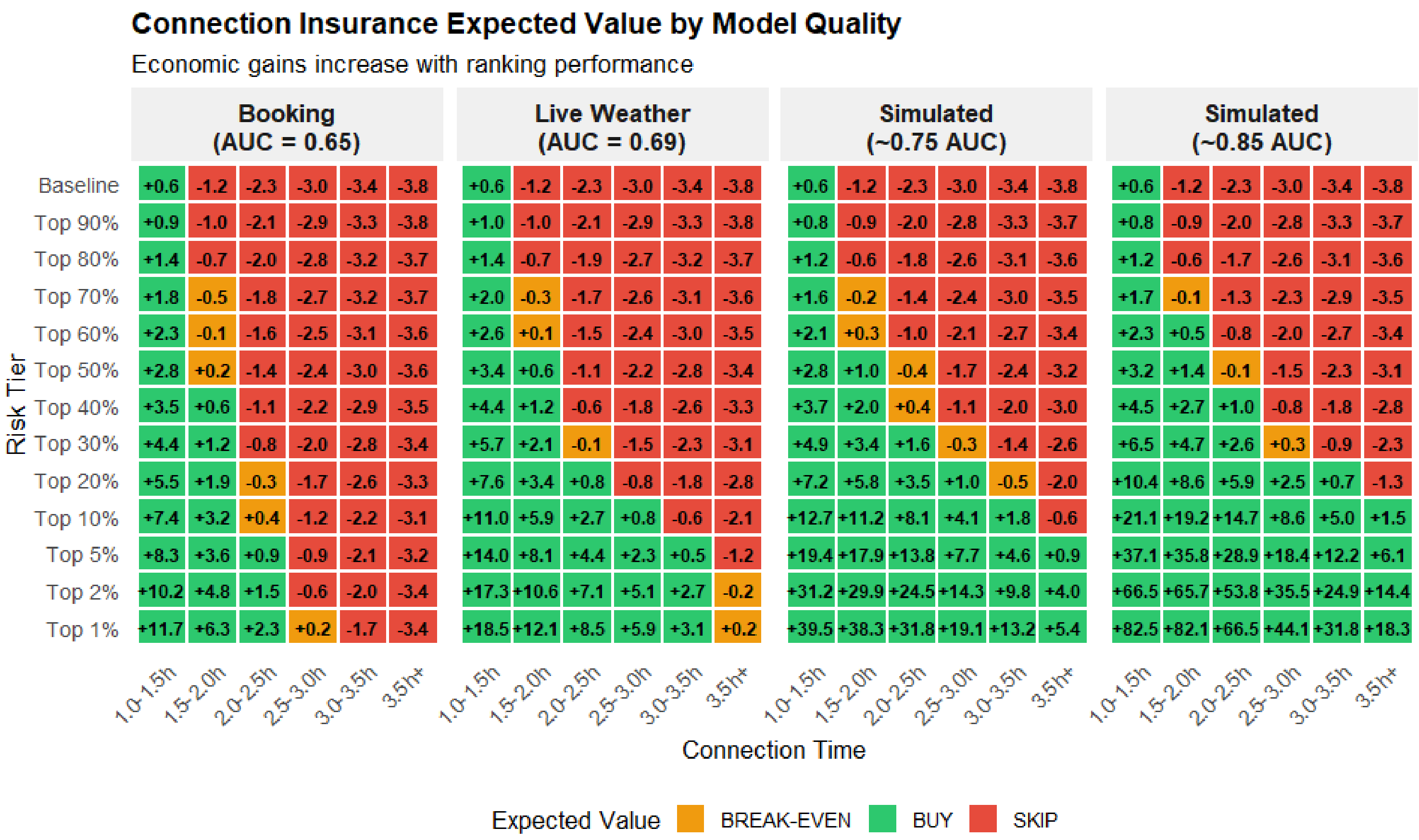

Figure 11: *Connection Insurance Expected Value (EV) Simulation Analysis. This matrix displays how the net economic return (Miss Rate - 4.5%) scales across risk percentiles and connection windows as model accuracy (AUC) improves, with high-performance simulations demonstrating massive value concentration reaching +82.5% in critical tiers.*

Connection Insurance: Effect of Model Quality

A similar pattern is observed for connection insurance. As model quality improves, miss probabilities become increasingly concentrated in the highest-risk groups, leading to an expansion of the "BUY" region as higher-quality models identify additional moderately risky flights as economically viable.

Short Connection Windows

For short connection windows (1 to 2 hours), the decision boundary changes only marginally across models. Even lower-quality models already identify the most relevant risk in this range, as moderate delays are sufficient to cause missed connections. Improvements in predictive performance therefore mainly increase expected value within already high-risk groups rather than expanding the set of flights for which insurance is recommended.

Long Connection Windows

For longer connection windows, the pattern differs. Here, missed connections are driven by relatively rare and severe delays, making accurate ranking more important. As model quality improves, the decision boundary expands substantially, with higher-quality models identifying additional flights where insurance becomes economically justified.

Practical Implications

From a practical perspective, this indicates that moderate predictive performance is sufficient to support effective decision-making for short connection windows. Further improvements primarily increase the magnitude of returns, particularly in the upper risk tiers, rather than fundamentally changing which flights should be insured.

5.5 Practical Implementation (Decision-Support Application)

Decision-Support Application

To demonstrate practical applicability, we implement the expected value framework in an interactive, web-based decision-support application using R Shiny (see Appendix I for interface visuals). The application translates model-based risk predictions into transparent, economically grounded recommendations for individual travelers.

User Inputs and Feature Construction

The tool requires only basic itinerary information, including origin, destination, departure date and time, operating airline, ticket price, and connection duration. All additional features are constructed automatically. In particular, the application derives calendar variables and historical weather averages and formats them into the input structure required by the trained XGBoost models, abstracting from the complexity of the underlying models and allowing users to interact through a small set of intuitive inputs.

Prediction Pipeline

When a user requests a prediction, the application executes a sequence of steps. First, input features are passed through the trained models to generate disruption risk scores. These scores are mapped to percentile-based risk tiers using the empirical distribution of model outputs. For connection insurance, the assigned risk tier is combined with the user's connection time to retrieve the corresponding missed connection probability from the empirical delay distributions. For cancellation insurance, the application retrieves the average cancellation rate within the corresponding risk tier.

Economic Output and Break-Even Pricing

These probabilities are translated into economic outcomes. Using the user's ticket price, the application computes the expected value of purchasing insurance and compares it to the premium, producing both a decision recommendation and a continuous measure of expected value. The application additionally provides the implied break-even insurance price, indicating the maximum premium at which insurance represents a fair deal given the traveler's estimated risk. This threshold should be interpreted as the best available fair price given observable data and model quality.

Interpretability via SHAP Values

To improve interpretability, the application uses SHAP values to summarize the drivers of predicted risk. Rather than presenting individual variables, feature contributions are aggregated into broader categories such as time and schedule, weather conditions, and route characteristics, providing users with a transparent explanation of why a given flight is classified as high or low risk.

Overall Contribution

The application operationalizes the core contribution of this paper by translating predictive risk estimates into individualized economic decisions. While insurance is not optimal for the majority of flights, the tool demonstrates that data-driven methods can identify the subset of travelers for whom insurance provides positive expected value, and determine the price at which such insurance becomes a fair deal.

6. Conclusion

Summary of Findings

This paper evaluates when travel insurance is economically justified at the time of booking, when most purchase decisions are made, by combining predictive models of delays and cancellations with an expected value framework. The results show that, at the booking stage, insurance is beneficial only for a relatively small subset of high-risk flights, implying that for most travelers, purchasing insurance is not economically optimal.

The Role of Information Timing

A central finding concerns the role of information. At the booking stage, predictive performance is limited and risk differentiation remains relatively weak. As additional information becomes available closer to departure, through realized weather conditions, the ability to identify high-risk flights improves substantially. This highlights that the value of insurance depends not only on underlying risk, but also on the timing and availability of information.

Cancellation versus Connection Insurance

The analysis further reveals an important distinction between insurance types. Connection insurance applies to a broader set of flights, as disruption risk can be meaningfully differentiated across a wider range of observations. In contrast, cancellation insurance is concentrated in a smaller high-risk segment, limiting its overall applicability. Nonetheless, even at the booking stage, the model provides value by helping travelers avoid particularly risky itineraries or identify cases where insurance may be justified.

Diminishing Returns to Predictive Performance

Improvements in predictive performance beyond the live-weather model primarily affect the concentration of risk in the upper tail of the distribution. While these improvements substantially increase expected value, especially among the highest-risk flights, they lead to relatively limited changes in the overall decision boundary. This suggests that better models mainly improve the efficiency of risk allocation rather than fundamentally altering which flights should be insured. The benefits of improved predictive performance are therefore asymmetric, accruing primarily through more precise risk targeting, while leaving individual insurance decisions largely unchanged.

Break-Even Pricing as a Practical Output

A key contribution of this paper is the translation of predicted risk into economically meaningful price thresholds. The framework provides a model-implied break-even price, representing the best available fair price given observable data and model quality. This allows travelers to evaluate whether an offered premium is favorable, rather than relying solely on binary purchase recommendations. Importantly, the results show that even with moderate predictive performance, these price thresholds remain informative and support economically grounded decision-making.

Decision-Support Tool

To demonstrate practical applicability, the framework is implemented in a decision-support tool that provides users with relative risk rankings and personalized break-even thresholds. This enables travelers to incorporate individual risk preferences while benefiting from model-based guidance.

Limitations

Several limitations should be noted. The analysis assumes risk-neutral behavior and focuses on expected value, while in practice travelers may exhibit risk aversion. In addition, connection risk is partly driven by flight cancellations, which are not directly captured within the delay-based framework used to model missed connections. This may contribute to the observed convergence in miss rates across risk groups at longer connection times, as delay propagation through connected flights means cancellation-driven disruptions are likely underrepresented in a delay-only framework (Schäfer et al., 2014). Furthermore, the modeling of weather conditions is subject to limitations in both spatial and temporal resolution. Weather data is matched to airports using nearby stations and aggregated at the daily level, which may not fully capture localized or intra-day variation in conditions that affect flight operations. In addition, the available time span of weather data limits the construction of historical weather features at the booking stage.

Future Research

Future research could extend the framework by incorporating behavioral preferences, explicitly accounting for cancellation-related disruption in connection outcomes, and leveraging longer historical datasets to improve the representation of expected weather conditions. Finally, while our framework identifies the actuarially fair break-even price for insurance, it does not account for the commercial markups applied by providers to cover administrative costs and profit margins. In practice, these loading factors would raise the market price above our estimated expected values, likely further narrowing the subset of flights for which insurance is an economically optimal choice for the traveler.

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Appendix A: Data Construction Details

Flight Performance Data

The Kaggle dataset covers departure delay, scheduled departure time, operating carrier, origin and destination airport, flight distance, and flight date among other variables.

Fare Data

Each row in the DB1B represents a unique itinerary defined by origin, destination, operating carrier, and fare, where a single row can cover multiple passengers using the same itinerary at the same price. After filtering on fare bounds and merging with the flight dataset, the DB1B yielded 5,947,884 unique route and quarter combinations, which were merged with the flight dataset at a 99.88% match rate, leaving 642,292 flight records before final weather merging.

Weather Data

Daily weather data was downloaded for all 370 stations from 2010 to 2019, yielding 1,338,479 observations, which falls 0.9% short of the theoretical maximum of 1,350,500 (370 stations × 365 days × 10 years), attributable to normal station downtime and data transmission gaps. The GUST variable was dropped due to 40% missing values. The following variables were retained: average temperature (TEMP), dew point (DEWP), visibility (VISIB), average wind speed (WDSP), maximum sustained wind speed (MXSPD), maximum temperature (MAX), minimum temperature (MIN), precipitation (PRCP), and snow depth (SNDP). Snow depth missing value codes (999.9) were replaced with zero, as missingness clusters in summer months and lower latitudes, indicating absence of snow rather than failed measurements. This is supported by geographic and seasonal validation plots shown in Figures A1 and A2. Visibility values were capped at 10 miles, consistent with standard aviation reporting conventions where 10 miles is the standard reporting ceiling.

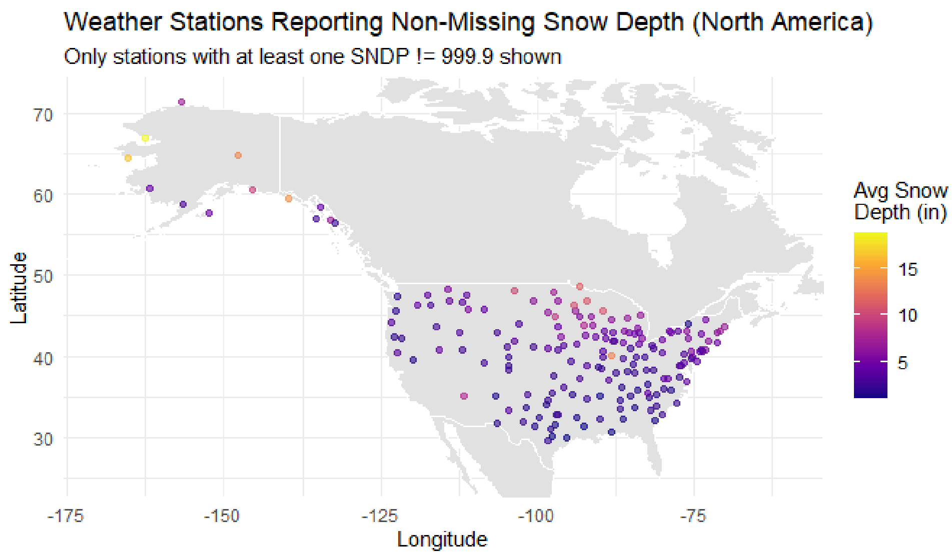


Figure A1: *Weather stations in North America reporting at least one non-missing snow depth value (SNDP ≠ 999.9) over 2010–2019. Colour indicates average snow depth in inches. Snow depth reports cluster in northern states, Alaska, and mountain regions, consistent with expected geographic patterns and supporting the validity of replacing 999.9 codes with zero.*

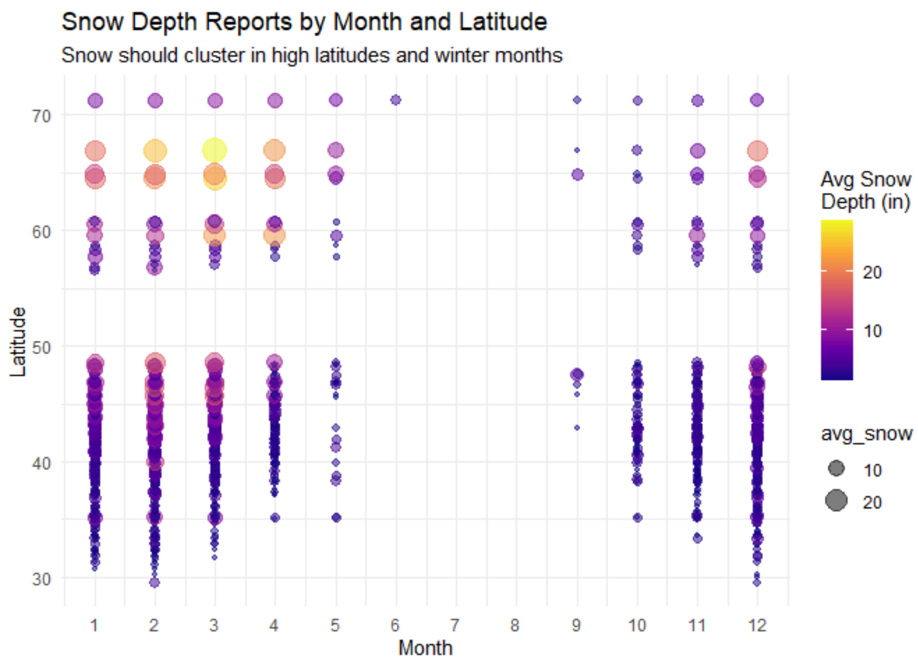


Figure A2: *Average snow depth by month and latitude for stations reporting at least one non-missing value. Snow depth concentrates in high latitudes and winter months (January–May, October–December), with a complete absence of non-missing values in June, July, and August across all latitudes, indicating that 999.9 codes in summer months reflect genuine absence of snow rather than measurement failure.*

Appendix B: Further motivation for use of XGBoost as our main Model

We first compared XGBoost with lasso to test for a significant improvement of predictive power. We tested lasso and XGBoost on depdelay probability, quantity of depdelay and cancellation probability. XGBoost clearly works better, likely due to the many non-linear patterns in the data. Due to this significantly better performance, we used XGBoost for the rest of our project, so also for outcome variables not tested with Lasso, arrdelay for example. Arrdelay wasn't tested initially, but we found it was more useful for creating an applicable insurance framework. Quantity of delay prediction performed extremely poorly in both models, so this was the reason we chose to model probabilities and rank these on risk.

We found and used XGBoost because it was a popular method in literature (e.g., , Gui et al., 2020; Chakrabarty et al., 2019). So, our methodology was planned around XGBoost, because it was such a frequently used and well-performing method in existing literature. However, after our project was done, we found some other algorithms that might have worked even better since our data contains many categorical and factor variables. These algorithms are CatBoost and Categorical XGBoost. They are also tree-based models, similar to XGBoost, but are designed in such a way to handle categorical data specifically well. We were curious how much better these algorithms performed and whether it was worth changing all our finished work over. Some models indeed performed better, but it was not such a significant increase that we deemed changing our main model as necessary.

Table 1: Departure Delay Prediction — Model Comparison

Model	AUC	Kappa	Precision	Recall	Specificity
Lasso / Logit Baseline (live weather)	0.6877	—	—	—	—
Normal XGBoost Baseline (live weather)	0.6937	0.201	29.8%	61.2%	67.0%
Categorical XGBoost Baseline (live weather)	0.6959	0.188	28.5%	67.8%	60.9%
Categorical XGBoost Tuned (live weather)	0.7000	0.197	29.2%	66.2%	63.2%
CatBoost Baseline (live weather)	0.6951	0.192	28.9%	65.4%	63.1%

Table 2: Cancellation Prediction — Model Comparison

Model	AUC	Kappa	Precision	Recall	F1
Lasso / Logit Baseline (live weather)	0.7758	—	—	—	—
Normal XGBoost Baseline (pure monthly)	0.6864	—	3.1%	63.4%	0.059
Normal XGBoost Baseline (hybrid weekly VISIB)	0.6829	—	3.0%	66.5%	0.057
Normal XGBoost Tuned (booking-time optimised)	0.6868	0.024	3.1%	63.7%	0.058
Normal XGBoost Baseline (live weather)	0.7820	0.079	6.0%	61.5%	0.110
Normal XGBoost Tuned (live weather optimised)	0.7892	0.073	5.7%	63.9%	0.104
Categorical XGBoost Baseline (live weather)	0.7871	0.070	5.5%	65.3%	0.102
Categorical XGBoost Tuned (live weather)	0.7976	0.071	5.5%	67.7%	0.103
CatBoost Baseline (live weather)	0.7641	0.059	4.9%	65.0%	0.091

Table 3: Arrival Delay Prediction — Model Comparison

Model	AUC	Kappa	Precision	Recall	Specificity	F1
Normal XGBoost Baseline (pure monthly)	0.6521	0.149	27.0%	60.2%	61.7%	0.373
Normal XGBoost Baseline (hybrid weekly VISIB)	0.6504	0.148	27.0%	59.7%	61.9%	0.372
Normal XGBoost Tuned (booking-time optimised)	0.6523	0.156	27.7%	57.1%	64.8%	0.373
Normal XGBoost Baseline (live weather)	0.6925	0.206	30.7%	60.7%	67.6%	0.408

Figure B1: *All models evaluated on test set (2019 data).*

Appendix C : Temporal Stability

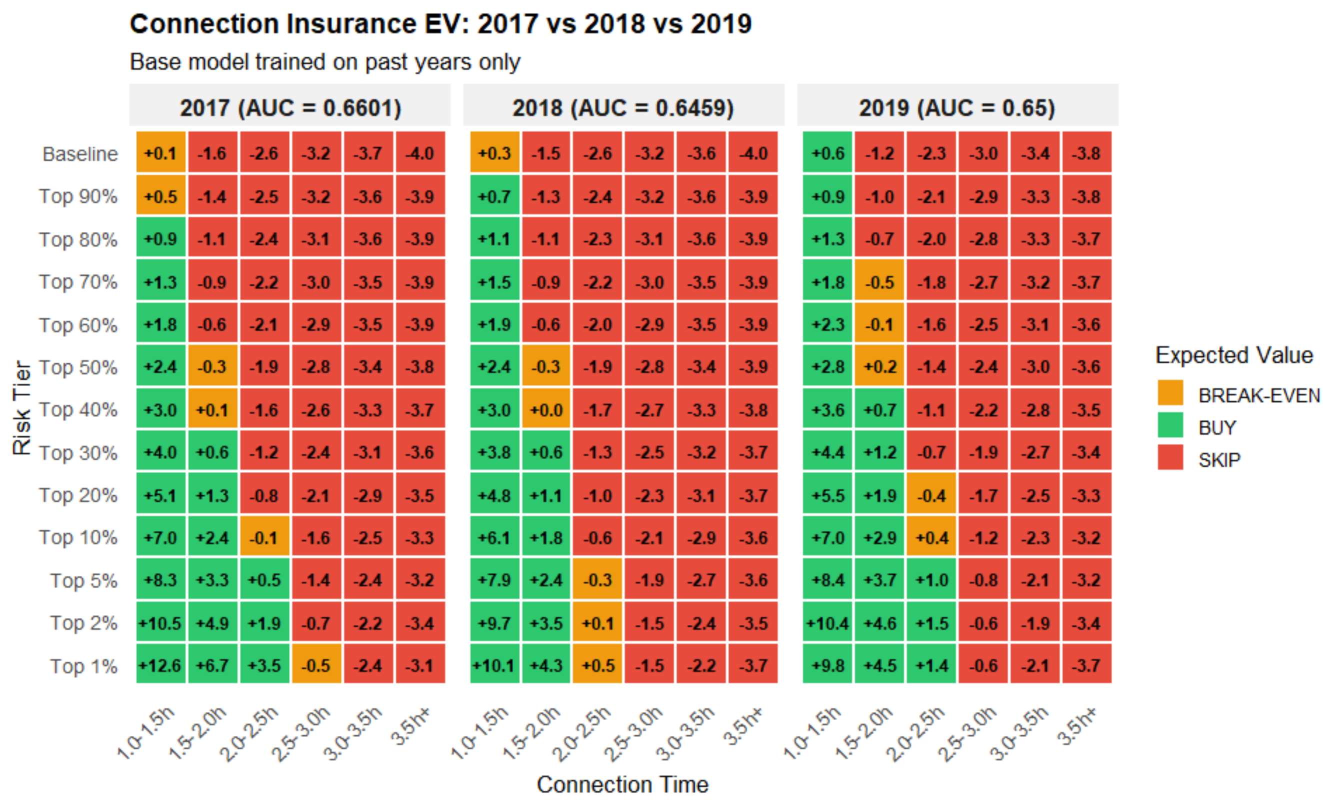


Figure C1: *Expected value (% of ticket price) of connection insurance across 13 risk tiers and six connection-time buckets for three out-of-sample test years (2017–2019), evaluated against a 4.5% break-even threshold.*

Temporal Stability of Model Performance

Risk Ranking Stability

We assess the temporal stability of the model's expected value (EV) and ranking performance by evaluating it across three out-of-sample years (2017–2019). Because severe delays are relatively rare events and subject to annual variation in underlying conditions, examining year-to-year stability is important for evaluating the model's practical applicability.

Decision Boundary Consistency

The results indicate that the categorical risk structure remains broadly consistent across years, while the underlying expected values exhibit some variation. This distinction has different implications depending on the use case.

From a traveler perspective, the model appears robust. Decision boundaries, defined by the break-even threshold, remain largely stable across years. As a result, the model continues to provide consistent directional guidance, even when underlying expected values fluctuate.

Tail Sensitivity

In contrast, variation in expected values is more pronounced within the highest-risk percentiles (e.g., Top 1% and 2%). Since these segments drive a large share of disruption risk, this suggests that expected values in the upper tail are sensitive to year-specific conditions.

Implications

Overall, the results indicate that while the ranking of risk is stable, expected value estimates in the extreme tail exhibit temporal variation. This suggests that applications relying on precise valuation, rather than ranking, may need to account for year-to-year fluctuations.

Appendix D : Regression Modeling

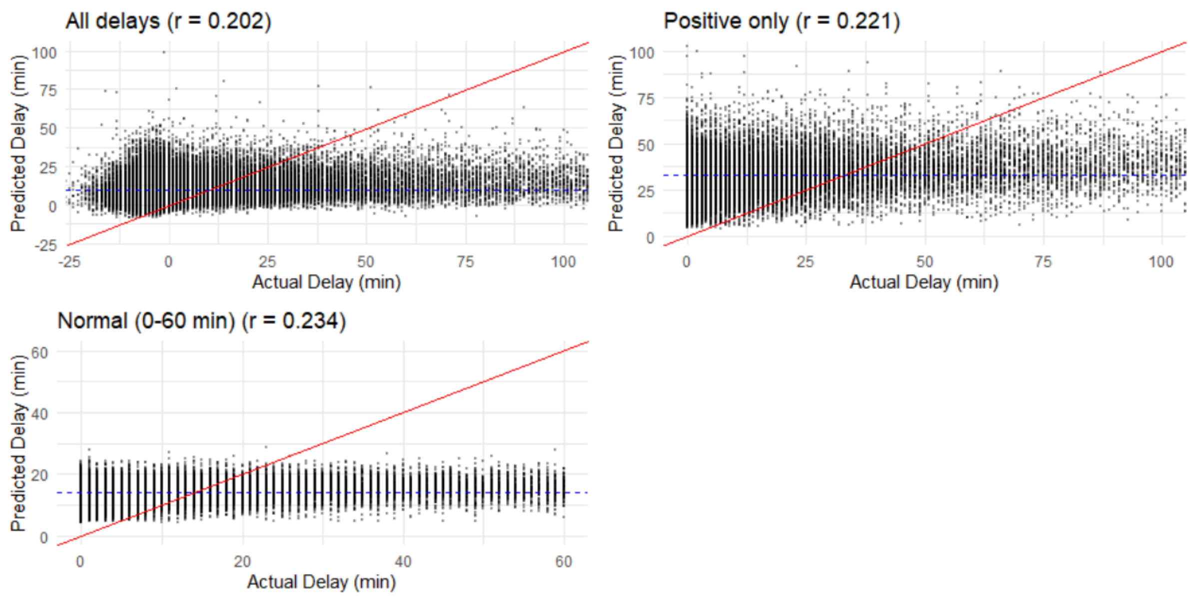


Figure D1: *Scatter plots of predicted versus actual arrival delay (minutes) for three subsets of the test set, illustrating the compression of predictions toward the mean and low correlation across all specifications (r = 0.20–0.23).*

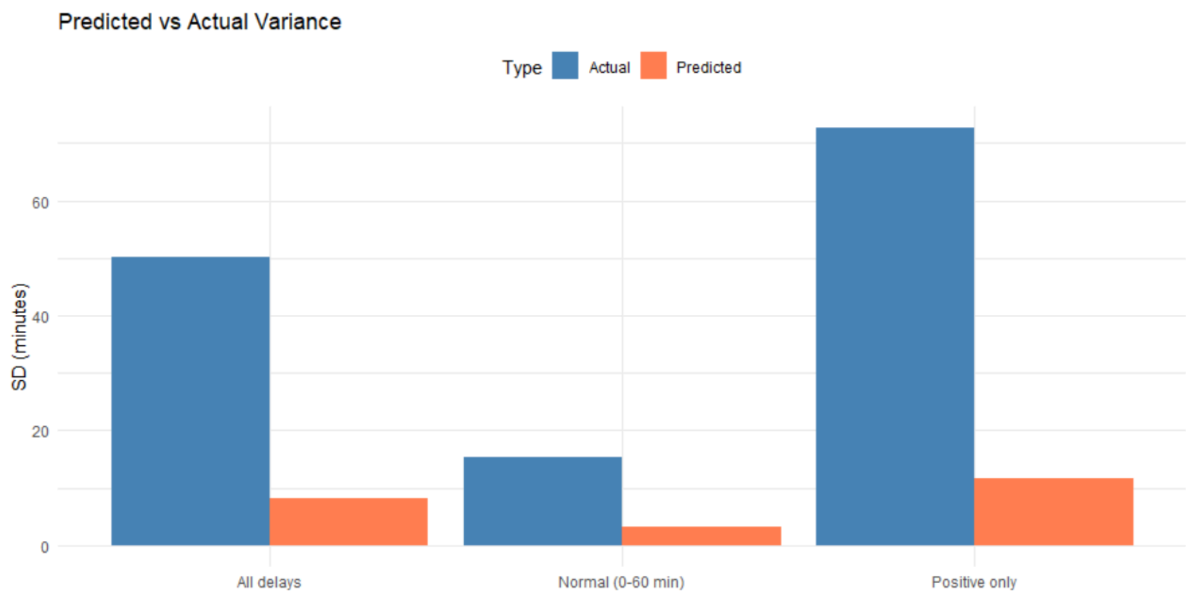


Figure D2: *Standard deviation of predicted versus actual arrival delays (minutes) across three subsets of the test set, showing that predicted delays exhibit substantially lower dispersion than realized delays in all cases, confirming the failure of the regression model to capture the heavy-tailed nature of delay distributions.*

Regression as an Alternative Approach

A natural alternative is to model delay magnitude directly using a regression framework. However, in this setting, regression models perform poorly due to the high variability and skewness of delay outcomes.

Prediction Compression and Correlation

As shown in Figure D1, predicted delays exhibit very low correlation with realized delays ($r \approx 0.20$–0.23) and are heavily compressed within a narrow range. In particular, the model fails to distinguish between moderate and severe delays, systematically shrinking predictions toward the mean. As a result, extreme delays, which are most relevant for missed connections, are not accurately identified.

Variance Comparison

This limitation is further reflected in the variance comparison, where predicted delays exhibit substantially lower dispersion than observed delays across all subsets. This indicates that the model fails to capture the heavy-tailed nature of delay distributions.

Implications for Insurance Decisions

This issue is critical in the context of insurance decisions, which depend on whether delays exceed specific thresholds rather than their exact magnitude. A regression approach therefore fails to capture the tail behavior that drives disruption risk.

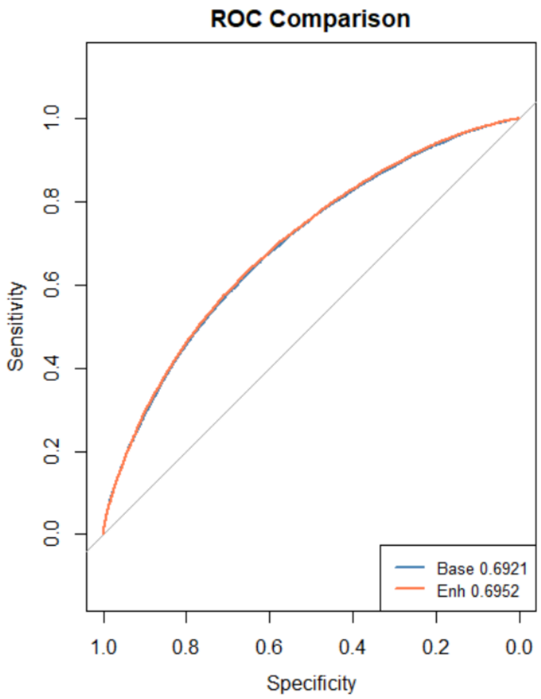
Classification Approach

For this reason, we adopt a classification-based approach (arrival delay ≥ 15 minutes), combined with empirical delay distributions within risk tiers. This allows us to model threshold exceedance directly while preserving information about delay severity where it is most relevant for decision-making.

Note on Departure Delays

We note that departure delays exhibit a much higher correlation with realized outcomes ($r \approx 0.95$). However, since arrival delays are the relevant outcome for missed connections, the limitations of regression in predicting arrival delay magnitude remain binding for the insurance decision problem considered here.

Appendix E : Cascade Modeling



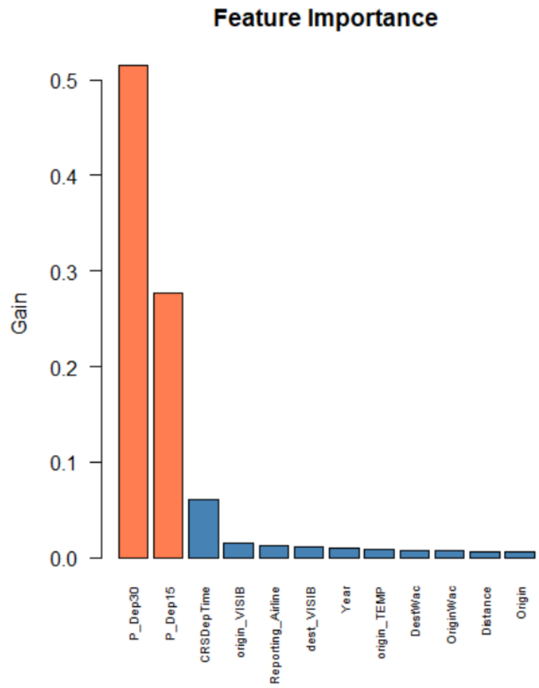


Figure E1 and E2: *Shows the AUC improvement and the feature importance, for live weather arrival cascade model using dep del > 15 and >30 as a predictor.*

Cascade Model Evaluation

We evaluate a cascade model that incorporates predicted departure delay probabilities (P_Dep15, P_Dep30) as additional features in the arrival delay model.

Marginal Performance Improvement

While departure and arrival delays are highly correlated (r = 0.96), the inclusion of these features results in only a marginal improvement in predictive performance (AUC: 0.6921 → 0.6955). Despite their high feature importance, the departure probability variables provide limited incremental information. This suggests that they primarily reflect the same underlying weather and operational conditions already captured by the baseline feature set, rather than introducing new predictive signals.

Model Selection

Given the small performance gain, the additional complexity of the cascade approach offers limited practical benefit. Furthermore, incorporating intermediate model outputs reduces transparency and interpretability, which is undesirable in the context of a decision-support tool. We therefore proceed with separate models for departure and arrival delay prediction.

Appendix F : Xgboost Importance Table

Full Feature Importance — Delay Model

Category	Feature	Booking (AUC = 0.65)	Live Weather (AUC = 0.69)
Airline	Reporting_Airline	0.0952	0.0907
Date/Time	CRSDepTime	0.3805	0.4542
Date/Time	Month	0.0775	0.0405
Date/Time	DayofMonth	0.0435	0.0102
Date/Time	DayOfWeek	0.0326	0.0111
Date/Time	Quarter	0.0033	0.0003
Distance/Duration	Distance	0.0159	0.0054
Distance/Duration	CRSElapsedTime	0.0139	0.0087
Origin/Dest	OriginWac	0.0465	0.0453
Origin/Dest	DestWac	0.0431	0.0271
Origin/Dest	Origin	0.0240	0.0089
Origin/Dest	Dest	0.0168	0.0135
Price	avg_fare	0.0154	0.0040
Weather	origin_PRCP	0.0200	0.0165
Weather	origin_WDSP	0.0184	0.0039
Weather	origin_MAX	0.0169	0.0107
Weather	dest_DEWP	0.0156	0.0138
Weather	dest_MXSPD	0.0152	0.0211
Weather	origin_VISIB	0.0141	0.0697
Weather	origin_TEMP	0.0129	0.0087
Weather	dest_WDSP	0.0120	0.0047
Weather	origin_MXSPD	0.0120	0.0371
Weather	dest_PRCP	0.0102	0.0081
Weather	origin_DEWP	0.0087	0.0291
Weather	origin_MIN	0.0087	0.0041
Weather	dest_VISIB	0.0078	0.0442
Weather	dest_MAX	0.0059	0.0034
Weather	dest_TEMP	0.0055	0.0023
Weather	dest_MIN	0.0043	0.0010
Weather	origin_SNDP	0.0022	0.0016
Weather	dest_SNDP	0.0013	0.0000

Full Feature Importance — Cancellation Model

Category	Feature	Booking (AUC = 0.69)	Live Weather (AUC = 0.79)
Airline	Reporting_Airline	0.1232	0.0576
Date/Time	Month	0.0771	0.0165
Date/Time	DayofMonth	0.0655	0.0192
Date/Time	CRSDepTime	0.0454	0.0354
Date/Time	DayOfWeek	0.0208	0.0096
Date/Time	Quarter	0.0125	0.0079
Distance/Duration	Distance	0.0531	0.0418
Distance/Duration	CRSElapsedTime	0.0321	0.0239
Origin/Dest	DestWac	0.0606	0.0449
Origin/Dest	OriginWac	0.0431	0.0401
Origin/Dest	Dest	0.0172	0.0224
Origin/Dest	Origin	0.0139	0.0242
Price	avg_fare	0.0315	0.0265
Weather	origin_MIN	0.0407	0.0189
Weather	dest_MIN	0.0389	0.0204
Weather	dest_WDSP	0.0299	0.0333
Weather	origin_PRCP	0.0295	0.0200
Weather	origin_VISIB	0.0270	0.0939
Weather	origin_DEWP	0.0261	0.0357
Weather	origin_WDSP	0.0253	0.0324
Weather	dest_MXSPD	0.0243	0.0337
Weather	dest_VISIB	0.0240	0.0627
Weather	dest_PRCP	0.0230	0.0248
Weather	dest_DEWP	0.0206	0.0376
Weather	origin_MXSPD	0.0205	0.0504
Weather	origin_MAX	0.0174	0.0233
Weather	dest_TEMP	0.0154	0.0472
Weather	origin_TEMP	0.0142	0.0405
Weather	dest_MAX	0.0140	0.0251
Weather	origin_SNDP	0.0066	0.0134
Weather	dest_SNDP	0.0065	0.0165

Table F1: *Full Feature Importance by Model Specification*

This table reports disaggregated XGBoost gain scores for all individual features across the delay and cancellation models under booking-time and live weather specifications. Features are grouped into categories in the main text for interpretability; this table provides the complete view.

Consistent with the aggregated results, time-related variables dominate at the booking stage, while weather variables increase substantially in importance when real-time conditions are incorporated. Individual weather features contribute relatively small shares but jointly account for a large portion of model performance.

Appendix G: Temporal Resolution of Weather Features (Monthly vs. Hybrid Visibility Aggregation)

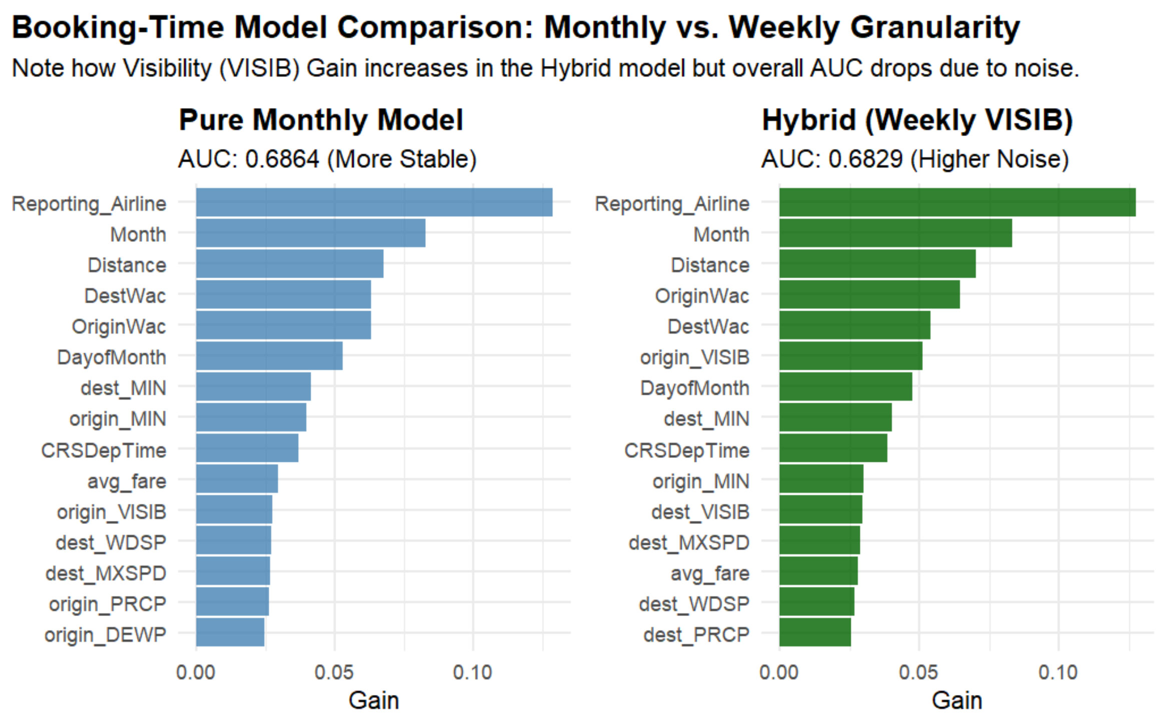


Figure G1*: Evaluation of Temporal Granularity in Weather Features*

Research Objective Following the observation in our "Live Weather" model (AUC: 0.79) that visibility is the primary meteorological driver for flight cancellations, we investigated whether increasing the temporal granularity of visibility data could improve the performance of our "Booking-Time" model. We compared a **Pure Monthly** specification (all weather features as monthly averages) against a **Hybrid** specification (visibility smoothed to 7-day weekly averages, other features as monthly averages).

Results: The "Feature Importance vs. Feature Quality" Trade-off The comparison revealed that while the model was sensitive to the more granular data, it did not lead to better predictive outcomes:

- **Pure Monthly Model (AUC: 0.6864):** In this baseline, visibility ranked 11th in feature importance (Gain ≈ 0.030). The model relied on the "Month" feature (ranked 2nd, Gain ≈ 0.085) as a stable proxy for seasonal weather patterns.

- **Hybrid Model (AUC: 0.6829):** Incorporating weekly visibility caused it to jump to 6th place in feature importance (Gain ≈ 0.050). However, this increased reliance resulted in a performance drop of -0.0035 AUC.

Analysis of Model Degradation The decline in performance is attributed to **sample sparsity**. While monthly aggregation provides stable estimates derived from over 240 observations per airport-month, the weekly window reduces this to approximately 56 observations per airport-week. Consequently, the weekly visibility estimates became "noisy." The model assigned more weight (Gain) to the visibility feature, but because that information was less reliable than the monthly averages, the overall prediction accuracy suffered.

Conclusion for Final Model Selection For booking-time prediction, we conclude that monthly weather aggregation is optimal. Attempting to refine temporal granularity (weekly visibility) introduced more noise than signal, leading to overfitting on sparse estimates. The Pure Monthly model's reliance on stable seasonal proxies ensures a more robust risk assessment for long-term booking horizons.

Appendix H: Simulation Methodology for Sensitivity Analysis

H.1 Objective

To quantify the potential economic value of future improvements in predictive accuracy, we conducted a sensitivity analysis using simulated prediction sets. These benchmarks represent "next-generation" performance levels, allowing us to evaluate how the Expected Value (EV) framework scales as models become more discriminative.

H.2 Theoretical Framework: Signal Detection Theory

The simulations are grounded in the equal-variance normal model of **Signal Detection Theory (SDT)** (Green & Swets, 1966). This framework models a classifier's output as a decision process that must distinguish between a "Signal" (disruption event) and "Noise" (non-event). The ability of the model to separate these two states is measured by the discriminability index, d' **(d-prime)**, which represents the distance between the means of the two distributions.

H.3 Simulation Logic

We generated latent decision values (L) by combining a binary signal (S) with random Gaussian noise (ε). The discriminability index (ε) was calibrated to targeted AUC benchmarks using the standard transformation $AUC = \Phi(d'/\sqrt{2})$, as detailed in Macmillan & Creelman (2004). This ensures the simulated models maintain the same probabilistic characteristics as real-world tree-based classifiers.

For each observation, the latent evidence is calculated as:

$$L = \varepsilon + (S * d') - d'/2$$

To transform these values into the probability format required by the insurance framework, we applied the logistic function:

$$P(x) = 1 / (1 + e^{\wedge} - L)$$

H.4 Targeted Performance Benchmarks

The d values were calibrated to match the specific AUC targets used in our analysis. Higher d values represent models with a higher signal-to-noise ratio:

Model Category	**Target AUC**	**Discriminability** d'	**Rationale**
Cancellation	0.85	2.2	High-information weather forecasting
Cancellation	0.90	3.2	Near-perfect categorical separation.
Arrival Delay	0.75	1.5	Incremental improvement over baseline.
Arrival Delay	0.85	2.5	Advanced real-time structural data.

Appendix I : How the App works

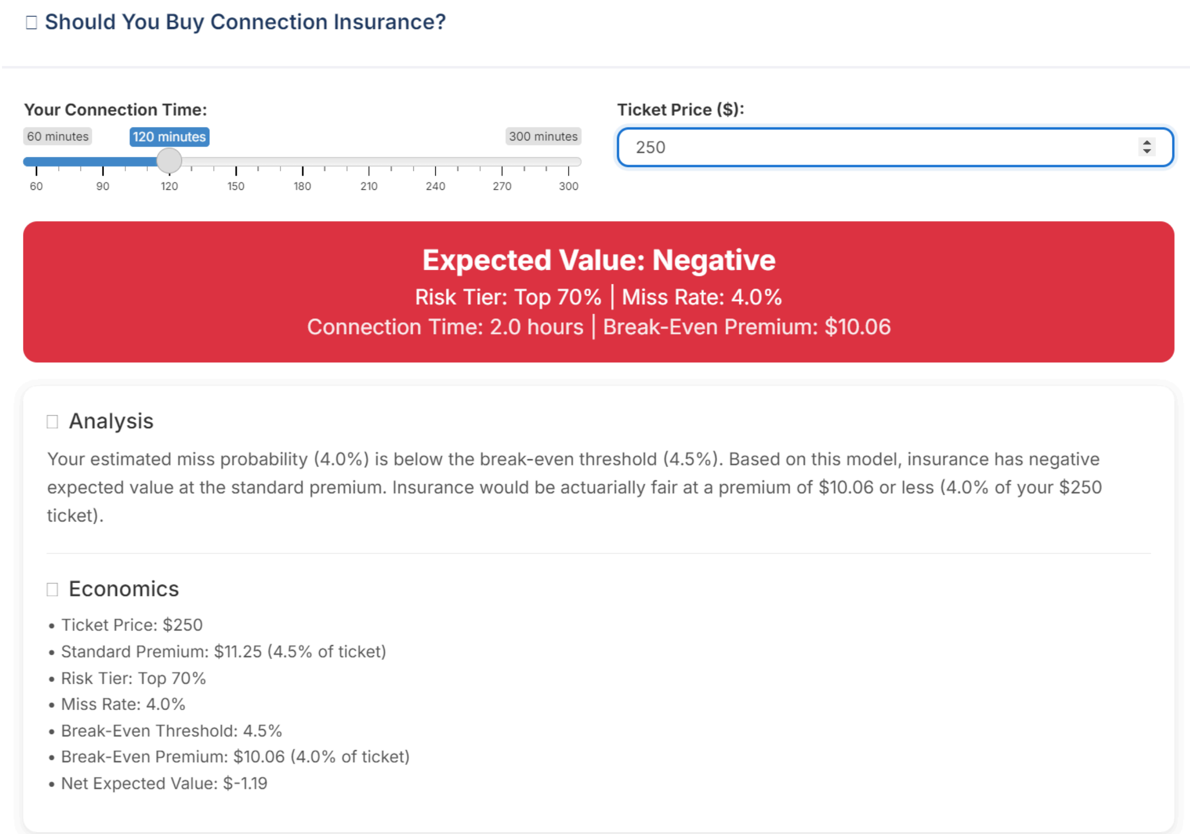


Figure I1: *Connection insurance decision output for an illustrative flight with a 2-hour connection window, showing the model-assigned risk tier, empirical miss probability, break-even premium, and net expected value based on a $250 ticket price.*

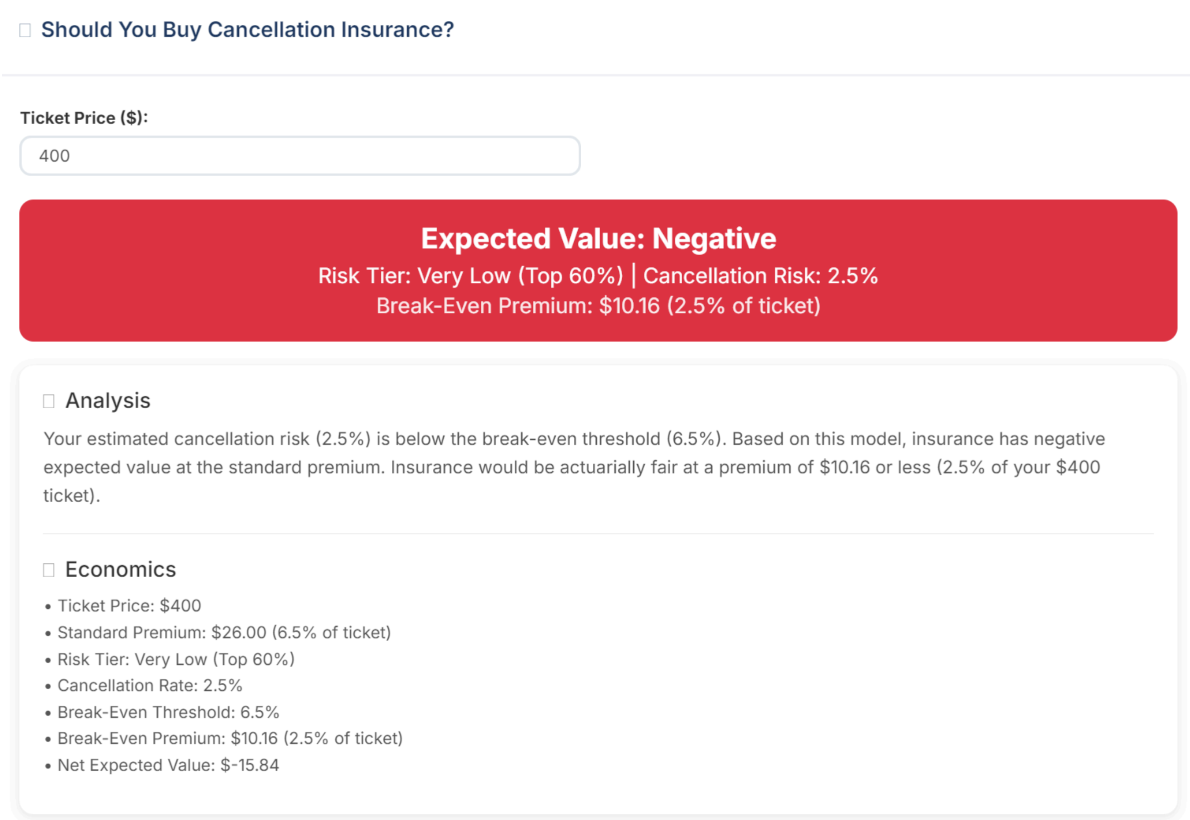


Figure I2:. *Cancellation insurance decision output for an illustrative flight, showing the model-assigned risk tier, empirical cancellation rate, break-even premium, and net expected value based on a $400 ticket price.*

Appendix J: Descriptive Data Analysis

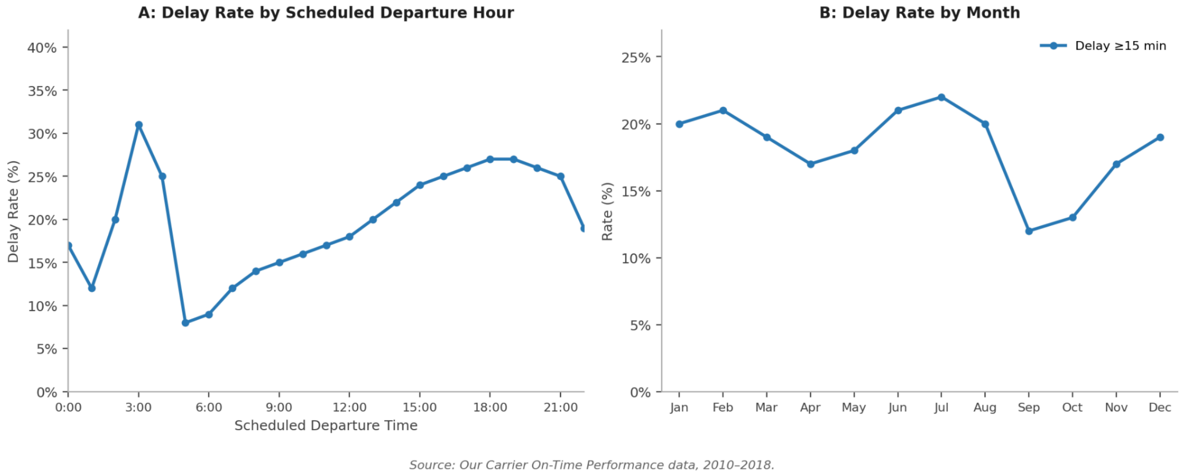


Figure J1: *Temporal structure of delay rates. Delay rates = share of flights with arrival delay ≥ 15 minutes. Sample restricted to 2010-2018 data.*

Table 1: Delay Rate by Airline (Top 10 by Delay Rate)

Airline Code	Delay Rate	No. of Flights
B6	24.2%	25,779
F9	23.6%	8,829
EV	22.1%	43,649
G4	22.0%	2,005
NK	21.5%	7,911
XE	21.5%	7,821
MQ	21.1%	30,240
OH	21.1%	7,048
VX	21.1%	3,954
AA	19.9%	70,929

Figure J2: *Showcase of structural difference in arrival delay rates regarding airlines, even in the top 10. Sample restricted to 2010-2018 data.*

Table 2: Delay Rate by Origin Airport (Top 10 by Delay Rate)

Origin Airport	Delay Rate	No. of Flights
EWR	25.5%	11,910
ORD	24.2%	29,674
SFO	23.2%	16,385
LGA	22.8%	11,377
DFW	22.1%	25,688
JFK	21.9%	10,657
DEN	21.2%	21,047
BOS	20.6%	12,181
BWI	20.4%	10,211
CLT	20.3%	14,906

Figure J3: *Showcase of structural difference in departure delay rates regarding airports, even in the top 10. Sample restricted to 2010-2018 data.*